

Aggregate Effects of Public Health Insurance Expansion: The Role of Delayed Medical Care

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October 2025

Abstract

Many older adults in the United States delay medical treatment until they qualify for Medicare at age 65. This can greatly increase their healthcare costs as they forgo cheaper earlier treatment, replacing it with more expensive care later. This paper studies the aggregate implications of delayed care focusing on the effects of expanding access to public health insurance before age 65. I build an overlapping generations model with rich household heterogeneity that is estimated to match quasi-experimental evidence on delayed care. My results indicate that the cost-savings are small, and expansion is unlikely to pay for much of its cost.

I thank Titan Alon, Mariacristina De Nardi, Roozbeh Hosseini, Minki Kim, Karen Kopecky, David Lagakos, Vegard Nygaard, Valerie Ramey, Kai Zhao, and various seminar participants at the SED annual meeting in Cartagena, the Federal Reserve Bank of Minneapolis, and UCSD for their helpful comments. Any errors, of course, remain my own. Mitchell VanVuren: Vanderbilt University email: mitchell.vanvuren@vanderbilt.edu

1. Introduction

It has long been said that an ounce of prevention is worth a pound of cure. The notion has led many — including health researchers ([Galvani et al. 2020](#), [Zieff et al. 2020](#)), policymakers ([WSJ 2023](#)), and economists ([Ozkan 2014](#)) — to argue that expanding public health insurance and access to preventative care can pay for some or all of its own cost as earlier treatment is cheaper. Intuitively, the potential savings seem large. A mild case of coronary artery disease (the build-up of plaque in one’s arteries) can be treated at relatively low cost — atorvastatin (a common generic anti-cholesterol medication) costs only \$20 per month. However, if a mild case worsens due to an uninsured patient delaying care, it can require surgical treatment (bypass surgery) which carries an average cost of \$169,000 ([Benjamin et al. 2018](#)). Providing an individual with health insurance can encourage earlier treatment, leading to an overall decline in outlays.

In the United States, this problem is particularly stark for individuals approaching age 65, the qualification threshold for Medicare. There is substantial evidence that individuals put off healthcare until immediately after this threshold, including discrete jumps at age 65 in the usage of screening services ([McWilliams et al. 2003](#)), cancer diagnoses ([Patel et al. 2021](#)), and many major procedures ([Card et al. 2008](#)). Further, many medical studies support the intuition above and show that delaying treatment leads to higher eventual treatment costs as disease progresses (e.g. [Gehi et al. 2007](#), [Herkert et al. 2019](#), [Fukuda & Mizobe 2017](#)).

In this paper, I study the consequences of public health insurance expansion in the spirit of the macro literature on health (e.g. [De Nardi, French & Jones 2016](#)), with a focus on the long-run cost savings that arise from reductions in delayed care. I construct a heterogeneous agent overlapping generations model featuring health investment and endogenous mortality. Following much of the literature, individuals build and maintain health capital, which reduces mortality risk, through medical spending each period (see [Fang & Krueger 2021](#), for an overview). Individuals face a choice between purchasing health insurance or not which leaves some low-income individuals uninsured until they receive Medicare at age 65.

For uninsured individuals approaching age 65, the optimal strategy is one of delaying healthcare spending; they treat their health as an asset and substitute consumption for medical care, running down their health capital. After turning 65 and gaining health insurance, these individuals compensate for their period of low spending through higher use of medical care which is costly for the government as these individuals are now covered by Medicare. Because it is more cost-effective to maintain a high level of health and treat disease early, there are indeed long-run cost savings from pro-

viding these individuals with health insurance and reducing the extent to which they delay care.

The key question is the extent to which the cost-savings from reductions in delayed care help offset the overall cost of expanding public health insurance access. This is a quantitative question, and to discipline the model I use two key pieces of evidence. The first, motivated by previous papers studying healthcare utilization around Medicare qualification, is the jump in total healthcare utilization that occurs when individuals turn age 65. I measure this using the Medical Expenditure Panel Survey and find evidence consistent with a 1 percent jump in total utilization. Matching this moment in the model disciplines the intertemporal elasticity of substitution of healthcare spending in the health production function (if this elasticity is large, individuals will be very willing to delay care as it carries little cost and vice versa) which determines the extent to which reductions in delayed care lead to overall cost savings.

Second, I leverage the quasi-experimental evidence of [Miller et al. \(2021\)](#), who use a diff-in-diff framework to show that expansion of Medicaid (the US's public insurance program for poor individuals) reduced mortality for low-income near-old (ages 55-64) individuals by 0.14 percentage points. This pins down the overall productivity of the health production function (if productivity is high, gaining insurance leads to a large decline in mortality and vice versa) and determines the extent to which reductions in delayed care lead to reductions in mortality.

The estimated model accurately reproduces many untargeted moments related to health and the expansion of public health insurance. Most importantly, it closely matches the strong left-skew exhibited by the distribution of health — many individuals are in perfect or close-to-perfect health while a few individuals are extremely sick — while only the average level of health is included as an estimation target. The model also replicates the longer-run dynamics of the decline in mortality following Medicaid expansion, predicting a small decline immediately after expansion that grows by a factor of 3x after 4 years, almost identical to the result observed in data, despite these dynamics not being targeted in estimation. Finally, the model closely reproduces an array of untargeted aggregate statistics such as the overall cost of public insurance expansion under the 2014 ACA, the percentage of the population covered by public insurance before and after expansion, and variation in life expectancy with respect to income.

Using the estimated model to evaluate the impact of an expansion of public health insurance similar to the expansion of Medicaid under the 2014 ACA yields more pessimistic results than the intuition from the comparison between atorvastatin and bypass surgery might suggest. Overall Medicare outlays fall by only \$8 for every \$100 spent on

expansion, suggesting that there is at best modest scope for reductions in delayed care to offset the cost of public health insurance expansion.

The model suggests two primary reasons for this result. The first arises from endogenous mortality. The decline in mortality raises the total number of people older than 65, increasing Medicare outlays. Although this decline represents an improvement in welfare, it pushes against (and even threatens to undo) the reduction in costs. In a counterfactual model in which mortality is exogenous and, consequently, this increase in costs is zero by construction, the cost savings due to expansion are almost four times as large (\$31 per \$100 spent), suggesting that it is this countervailing force that leads the model to pessimistic predictions, rather than the fact that delayed care is unimportant.

The second reason for the small cost-savings is that the expansion is a blunt policy, resulting in poor targeting (at least from the perspective of addressing delayed care). First, although low-income individuals (those targeted by expansion) do exhibit substantial delay of care due to higher rates of uninsurance, even median-income individuals substantially delay care as uninsurance rates are still fairly high in this group. Second, the model suggests that delay among the young has only small consequences for their long-run health, and a generic expansion directs many resources towards this care. An alternative policy that directly targets delayed care among the near-elderly by easing uninsured individuals into coverage as they approach age 65 can achieve substantially better cost-effectiveness (reducing Medicare outlays by \$93 for every \$100 spent) and almost pays for itself.

Despite limit scope for cost savings and reductions in delayed care, expansion is somewhat effective at reducing mortality and, consequently, increasing welfare for those who receive access. The probability of surviving to age 70 conditional on reaching age 60 (i.e. across the age 65 threshold around which delayed care is most severe) increases by about 1.2 percentage points, and welfare for those who receive access to Medicaid increases by 11 to 13 percent of consumption with the reduction in mortality accounting for a substantial portion (about 25 percent) of these gains. Although the remaining individuals are worse-off in general equilibrium due to higher taxes, the overall (utilitarian) impact on welfare is positive, increasing by 1 percent.

Overall, these results suggest that the cost reductions stemming from reduced delayed care do not alone provide justification for the expansion of income-based public insurance programs. Although the expansion does reduce mortality and substantially increase welfare for those impacted, these gains should be interpreted as largely the result of redistribution rather than any overall improvement in cost-effectiveness — nearly the full cost of the policy must be borne by taxpayers. In essence, the model

suggests that delayed care does not present a “free lunch” in terms of the expansion of public insurance.

Related Literature: This paper is inspired by and builds on a growing macroeconomic literature evaluating the role of public health insurance using macroeconomic models. [De Nardi, French & Jones \(2016\)](#) evaluate Medicaid in the context of late-in-life insurance and find that it is approximately the correct size. [Aizawa & Fu \(2020\)](#) examine the interaction between risk-pool cross subsidization and Medicaid expansion and find that expansion leads to higher welfare gains. [Kopecky & Koreshkova \(2014\)](#) examine the impact of health insurance and late-in-life medical expenses on wealth accumulation. [Pashchenko & Porapakkarm \(2013\)](#) evaluate whether the welfare gains from expansion come from primarily regulatory changes or primarily redistribution and find that the welfare gains overwhelming come from the latter. [Jung & Tran \(2016\)](#) examine this same question in a more complex model with endogenous health expenditure and find a similar answer. To measure health, I leverage the approach of [Hosseini, Kopecky & Zhao \(2021\)](#) by constructing frailty indices.

This paper also contributes to the literature on macroeconomics and insurance such as [Kaplan & Violante \(2010\)](#). I add to a large and growing literature on self-insurance in two-asset heterogeneous agent models such as [Kaplan, Moll & Violante \(2018\)](#) since health functions as an asset in the model. In a similar vein, health in the model can also be thought of as a durable good as in [McKay & Wieland \(2019\)](#).

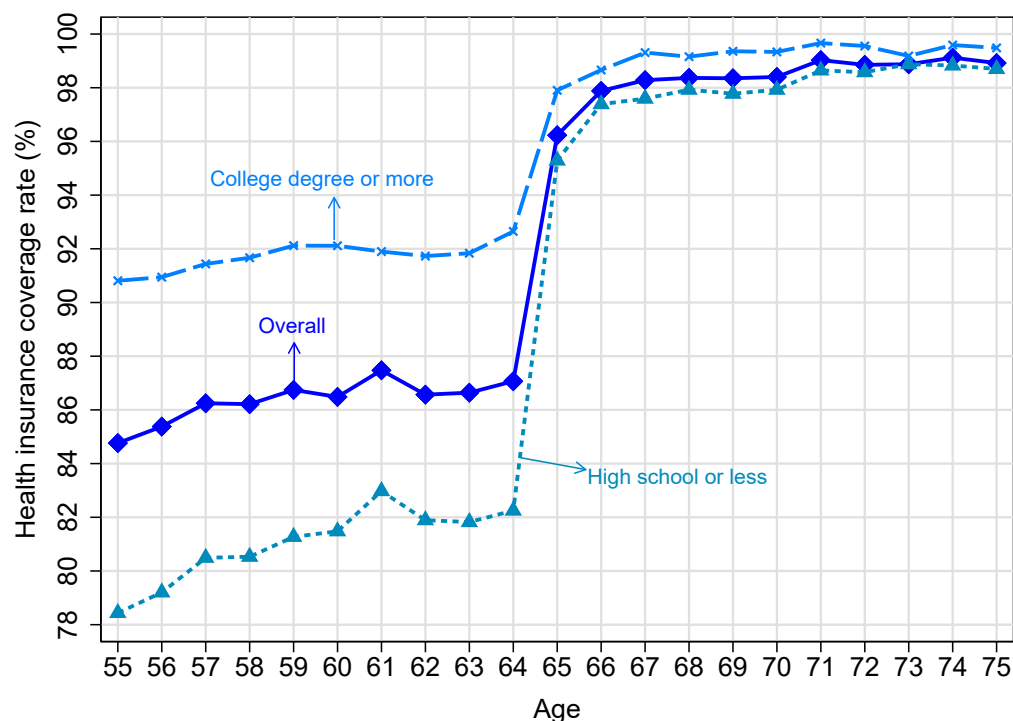
This paper is most closely related to the work of [Ozkan \(2014\)](#) which estimates a macroeconomic model of health spending and argues that shorter optimal lifespans for poorer individuals cause these individuals to under-spend on preventative care early in life, face a more costly distribution of late-in-life health shocks, and spend more on healthcare overall. This paper, instead of focusing on early-in-life preventative care, focuses on delay that occurs near the Medicare age threshold. In contrast to [Ozkan \(2014\)](#), I find that cost-savings from expanding access to care are small.

Finally, this paper also contributes to the broader literature on health and healthcare spending in macroeconomic models. [De Nardi, French & Jones \(2010\)](#) examine the role that late-in-life medical expenses play in individuals’ optimal savings behavior. [Cole, Kim & Krueger \(2019\)](#) estimate optimal insurance policy in a model where health and labor market risks are intertwined and health insurance induces a moral hazard inefficiency.

2. Some Facts on Healthcare Near Age 65

A remarkable feature of the US healthcare system is the discrete and sudden increase in health insurance coverage that occurs at age 65. After age 65, the government pro-

Figure 1: Health Insurance Coverage by Age and Education



Note: This figure displays the percentage of individual who self-report having health insurance coverage as a function of age and educational attainment. Calculated from NHIS data from 2002 to 2012.

vides nearly universal healthcare through Medicare while there is no universal form of coverage for those younger than 65. Figure 1 displays the rate of health insurance coverage as a function of age and education (used as a proxy for income that is not subject to selection concerns around the age 65 threshold due to wealthy individuals retiring or reducing working hours early and reporting low incomes near age 65) calculated from the National Health Interview Survey. Before age 65, there is a substantial gap in coverage between educational groups of roughly 10 percentage points; however, at age 65 there is a jump in coverage for both education groups and a large convergence in coverage rates due to the sudden availability of Medicare.

This increase in insurance coverage is both quantitatively large and extremely salient — it is a well-known fact among US individuals that Medicare eligibility begins at age 65. As a result, we might expect to see significant changes in behavior around the age 65 threshold, particularly for individuals who have no health insurance and anticipate large declines in the marginal cost of receiving healthcare upon turning 65. While this is not the first paper to notice this fact (e.g. [McWilliams et al. 2003](#), [Card et al. 2008](#)),

here I reexamine this phenomenon using nationally representative data that (ostensibly) cover all healthcare spending. I focus in particular on heterogeneity with respect to insurance status before turning 65 with an eye towards confirming the intuition that jumps in expenditure should be largest for those without coverage and, consequently, that expansion of public insurance can be used to mitigate this delay.

2.1. Data

I use the Medical Expenditure Panel Survey (MEPS), which will also eventually serve as the primary source of data for quantification of the structural model. These nationally-representative data contain detailed information on individual healthcare expenditure, insurance coverage and plan details, and health status. The panel structure of the MEPS allows observation of individuals' eventual health outcomes (at a time horizon of up to two years, after which they drop out of the rolling panel) conditional on their characteristics such as health or age, which will eventually be important in providing discipline to the model.

A second useful feature of the MEPS is that the data on healthcare expenditures are largely measured using billing data from healthcare providers, rather than estimates from surveyed individuals.¹ Thus these data reflect actual payments made to providers and are not subject to rounding bias (as is common in survey data) or issues of nominally inflated medical bills that are later reimbursed by insurance companies at lower rates, though both the amount initially charged ("Charges") and the amount eventually paid ("Expenditures") are recorded, allowing one to directly observe this.

In order to study a potential jump in healthcare utilization when individuals turn 65, I leverage the fact that the Event-level MEPS files record the month in which any instance of medical expenditure occurs. Combined with information on individuals' months-of-birth (available in the Full-Year files), I am able to measure the timing of expenditure relative to when an individual turns 65 reasonably precisely up to the month before they turn 65 and the month after. Of course, observing only months rather than precise dates means that expenditure that occurs in the month that an individual turns 65 cannot be unambiguously classified as occurring either before or after Medicare eligibility.

I consider three categories of expenditure, measured multiple ways for a total of

¹Unfortunately, not all spending from all individuals is taken from billing data. Whether billing data is collected for a health event depends on the insurance coverage of the individual and the provider seen. Appendix Table A.1 summarizes the fraction of spending taken from billing data for two key categories of provider: office-based physicians and hospitals. The MEPS attempts to use information from the surveyed providers to impute individual-reported spending at providers who are not surveyed. See [Cohen \(2003\)](#) and [Zuvekas & Olin \(2009\)](#) for more details.

eight measures of utilization. The categories I consider are Office-Based Medical Provider Visits (which correspond to typical instances of healthcare such as routine checkups, basic in-office treatment, and even some simple procedures such as biopsies), Hospital Inpatient Visits (which correspond to more serious treatments, procedures, and surgeries that require admission to a hospital), and Other (which includes Outpatient Visits, Emergency Room Visits, and Home Health).²

In each category, I measure utilization in (up to) three ways that all have unique advantages and disadvantages. The first is directly via quantity consumed (i.e. ignoring prices). For office visits, this simply counts the number of visits and, for hospital stays, the number of stays. For the "Other" category, I exclude this measure as the disparate types of expenditure in this category means quantities cannot simply be totaled up. The advantage of this measure is that it corresponds most closely to "real" utilization and does not reflect any changes in billing practices or the price of services that may occur when switching to Medicare. The disadvantage is that it does not measure quality — an office visit in which an individual declines a costly blood test is counted the same as a visit in which they receive the test. The second measure of utilization uses the total amount charged for a visit/stay/event. This has the advantage of measuring quality, but its disadvantage arises from the fact that bills are often nominally inflated only to later be reimbursed at lower rates by insurance. The final measure of utilization uses the total amount expended (i.e. actually paid) on a visit/stay/event, continuing to account for quality and overcoming this issue. This, however, introduces a new issue arising from the fact that different insurers may have different levels of negotiating power with providers and, in particular, that Medicare may leverage this to reimburse equivalent services at lower rates than private insurers. Due to the imperfections in each measure, I perform and report the analysis using all three measures, ensuring that my results reflect a robust notion of utilization.

2.2. Analysis

I begin by identifying individuals who turned 65 during their participation in the MEPS. As the results of this analysis will eventually be used to discipline a model of the pre-ACA-Medicaid-expansion United States, I use surveys from 2007 until 2013, the last full year before Medicaid expansion is implemented. I then consider the calendar year in which they turned 65 and compute all eight measures of healthcare utilization

²Notably, dental expenditures are excluded as are prescription medicines due to the fact that the MEPS records only the date of prescription and not the date of purchase, making it impossible to accurately classify the timing of expenditure.

(described above) for this year.³ Using this number, I compute for each month the percentage deviation in that month’s utilization relative to what would be expected if the total utilization for the year were evenly distributed. For example, if a person’s total expenditure on office-based visits was 1200 USD during the year, so that we would expect to see spending of 100 USD each month if it were evenly distributed, and they have an expenditure of 200 USD in June, I assign a value of 100 percent ($\frac{200-100}{100}$) to June.⁴ This is my outcome of interest that transparently measures the extent to which individuals skew their health utilization and spending towards certain months, potentially delaying care.

I construct these series for each of measure of healthcare usage and denote them $\text{PctDev_}X_{i,t}$ for measure X , person i , in month t . The estimating equation is then straightforward.

$$\text{PctDev_}X_{i,t} = \alpha + \beta_1 \text{Post_65}_{i,t} + \beta_2 \text{Turned_65}_{i,t} + \varepsilon_{i,t} \quad (1)$$

The variables $\text{Post_65}_{i,t}$ and $\text{Turned_65}_{i,t}$ are indicators representing whether a given month t occurred after or during the month individual i turned 65 respectively. As mentioned above, expenditure that occurs during the month an individual turned 65 could have occurred either before or after Medicare eligibility and thus is treated distinctly. Our coefficient of interest is β_1 , which measures the jump in healthcare utilization that occurs when an individual qualifies for Medicare.

2.3. Results

Table 1 displays the results of this exercise. Upon turning age 65, all three measures of utilization/expenditure for office-based visits increase. The number of visits increases by 1.7 percent, and the total value of charges and expenditures for these visits increase by 3.0 and 2.2 percent respectively. All three coefficients are statistically significant. In contrast, the three measures of utilization for hospital inpatient stays exhibit no such jump and are fairly precisely estimated null effects — the 95 percent confidence intervals rule out increases larger than 0.6 percent for all three measures. Other healthcare expenditures exhibit a similar null effect, though less precisely estimated.

To lend further credibility to these results, Appendix Figure A.1 displays the raw data by plotting the mean value of the outcome variable for all three measures of office-

³Calendar year rather than a relative measure (i.e. +/- 6 month after turning 65) is used in order to align with MEPS reference windows.

⁴Individuals with zero usage for the year, who otherwise induce a “zero over zero” problem, are manually assigned a value of 0 percent for each month under the intuition that their (lack of) utilization is perfectly evenly distributed, but all results below are robust to dropping these individuals entirely.

Table 1: Estimated Jump in Healthcare Utilization at Age 65

	Office-Based Visits			Hospital Inpatient Stays			Other	
	(1) Visits	(2) Charges	(3) Expend.	(4) Stays	(5) Charges	(6) Expend.	(7) Charges	(8) Expend.
Post-65	0.017*	0.030***	0.022*	-0.002	-0.002	-0.002	0.010	0.009
	(0.008)	(0.009)	(0.009)	(0.004)	(0.004)	(0.004)	(0.008)	(0.008)
Turned 65	0.017*	0.021**	0.018*	-0.007*	-0.007	-0.006	0.008	0.007
	(0.007)	(0.007)	(0.007)	(0.003)	(0.003)	(0.003)	(0.006)	(0.006)
Constant (Pre-65)	-0.011*	-0.017**	-0.013*	0.003	0.003	0.003	-0.006	-0.006
	(0.006)	(0.006)	(0.006)	(0.003)	(0.003)	(0.003)	(0.005)	(0.005)
<i>N</i>	5652	5652	5652	5652	5652	5652	5680	5680
Placebo p-value	0.173	0.346	0.110	0.950	0.928	0.927	0.428	0.625

Note: This table displays the regression coefficients for equation 1 on MEPS data from 2007 to 2013.

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

based visits (columns 1, 2, and 3) and all three measures of inpatient stays (columns 4, 5, and 6) as a function of the number of months until an individual turns 65. The jump in office-based visits is visually apparent, as is the lack of a jump in inpatient stays, so that the results pass the “eyeball test”. I also perform a placebo test where I use the month that an individual turns age 66, rather than 65, as the cutoff. At this threshold, there is no change in Medicare availability, and we would expect the results to disappear. The p-value for these tests are displayed in the final row of Table 1. As expected, none of the estimated coefficients are significant, assuaging concerns that these results are being driven by things other than insurance availability.

Heterogeneity: If the jump in healthcare usage upon turning 65 does indeed stem from the reduction in costs due to insurance coverage provided by Medicare, one would expect that this jump should be larger for individuals who were uninsured immediately before turning 65, as they experience the largest reduction in costs, and smaller for individuals who already had coverage before turning 65. Motivated by this fact, I partition individuals into three categories — those with private insurance coverage before age 65, those without insurance before age 65, and those with Medicaid coverage before age 65 — and re-run the analysis of spending on office-based visits separately for each category.

Table 2 displays the coefficient on interest, β_1 , for each combination of insurance type and measure of utilization. As expected, the estimated jump in spending is small and null for individuals with private insurance prior to turning 65. These people experience very little change in their cost of healthcare and thus face little incentive to delay care. On the other hand, the estimated jump for the uninsured is large and significant. The

Table 2: Estimated Jump in Office-Based Visits at Age 65 by Insurance Coverage

	(1)	(2)	(3)
	Visits	Charges	Expend.
Private	0.001 (0.011)	0.008 (0.012)	-0.001 (0.012)
Uninsured	0.096*** (0.019)	0.126*** (0.021)	0.106*** (0.021)
Medicaid	-0.034 (0.023)	-0.020 (0.026)	-0.010 (0.027)

This table displays the estimate jump after age 65 (β_1 in equation 1) in the three measures of office-based visit utilization based on individuals' insurance status immediately before turning 65. Note: Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

number of office-based visits increases by 9.6 percent and charges and expenditures on these visits increase by 12.6 percent and 10.6 percent respectively; these are the individuals who face the largest reduction in the cost of care and thus the largest incentive to delay. Interestingly, the estimated jump for those covered by Medicaid is also insignificant and, if anything, negative. This is a strong indicator that it is not necessarily low incomes that are driving the jump observed in the uninsured and is instead a result of (a lack of) insurance coverage specifically. It is also worth noting that the magnitude of this increase combined with Medicare's nominal 20 percent copayment rate implies an elasticity of spending (on office-based visits) to costs of about $-0.13 \left(\frac{0.106}{(0.2-1)/1} \right)$, a little smaller though still broadly consistent with the value of -0.2 measured in previous studies such as the RAND health experiment (Aron-Dine et al. 2013).

Summary: In short, there appears to be robust evidence that individuals are delaying healthcare until they turn 65 and receiving Medicare, after which their usage increases. The fact office-based visits, which are scheduled largely under the control of the patient and typically involve things like screenings and preventative care where there is fair amount of discretion in timing, increase while hospital inpatient stays, which involve more urgent matters that cannot as easily be put off, do not suggest that these results reflect the conscious delay of care by individuals rather than merely reflecting changes in billing practices. This is also underscored by the fact that this increase almost exclusively occurs among those without insurance prior to aging into Medicare. In a similar vein, those who had public health insurance coverage through Medicaid prior to turning 65 experience no such jump, suggesting that expanding the scope of public health insurance may be an effective tool in reducing the extent to which individuals delay care. These facts motivate a model of endogenous health expenditure and

credit-constrained individuals who (potentially) face incentives to delay care.

3. Model

The goal of the model is to allow the evaluation of the improvement in cost-effectiveness when reducing delayed care. Each period, individuals face a trade-off between consumption and investment into health. Insurance coverage, either purchased or provided by the government, reduces the marginal cost of health investment and of health emergencies. Because health and mortality occur at the individual level, I model the problem of individuals rather than of households. Although it lacks interesting behavior such as intra-household risk sharing, abstracting from household structure keeps the model tractable and creates a clear link between the model and health data, which are measured for individuals.

Time is discrete and runs infinitely. Individuals are heterogeneous in their income y , savings b , health h , and age a . Exogenous measure n individuals are born at age 18 each period and age increases by 1 each period thereafter. At the end of a period, an individual faces an age- and health- dependent probability of dying $\pi(h, a)$ (discussed in detail later).

3.1. Preferences

Individuals have preferences over streams of consumption $\{c_a\}_{a=18}^{100}$, labor supply $\{l_a\}_{a=18}^{100}$, and mortality risk $\{\pi_a\}_{a=18}^{100}$. I assume that individuals are born at age 18 and die with certainty at age 100 (i.e. $\pi_{100} = 1$). I follow the sparse literature on modeling preferences over mortality by having period felicity be equal to felicity from consumption and labor $u(c, l)$ plus a “joy-of-life” parameter $\bar{u}$ which represents the additional utility an individual receives simply for being alive for the period so that period felicity is given by $\bar{u} + u(c, l)$.

Individuals discount the future exponentially. Death, if it occurs, occurs at the end of each period so that π_a denotes the probability that an individual dies at the end of period in which they are age a and does not live to see age $a + 1$. From the perspective of today, an individual of age a sees their future felicity at age $a + 1$ as $\beta((1 - \pi_a)(\bar{u} + u(c_a, l_a)) + \pi_a \cdot 0)$ (the utility of death is normalized to zero). All together, an individual’s present-discounted lifetime utility is given by

$$U(c, l, \pi) = \sum_{a=18}^{100} \left[\prod_{j=17}^{a-1} (1 - \pi_j) \right] \beta^a [\bar{u} + u(c_a, l_a)] \quad (2)$$

In essence, period mortality risk π_a acts as a time-varying discount factor by inducing

individuals to put less weight on utility from periods that they are less likely to live to see.

3.2. Health and Healthcare Expenditure

An individual's health status is written as a health index h with higher values of h representing healthier individuals. In this way, h can be thought of as a sort of "health capital". When brought to the data, h will correspond to a measured health index that lies in $[0, 1]$ with $h = 1$ and $h = 0$ representing maximally and minimally healthy individuals respectively.

Individuals can increase their health status using medical care, labeled i for health investment, through the health accumulation equation

$$h_{t+1} = (1 - \delta_a - \delta_x)h_t + \phi i_t^\psi \quad (3)$$

where δ_a is the natural rate of health depreciation which may depend on age, δ_x is the (possibly zero) depreciation from an acute emergency shock (discussed in detail in two paragraphs), ϕ is a productivity parameter governing how effectively dollars of healthcare spending translate into units of health, and ψ is a returns-to-scale parameter that governs how quickly health can be accumulated. I allow δ_a to vary linearly with age, reflecting the fact that the process of health maintenance becomes more difficult with over time.⁵

This (non-emergency) medical care expenditure represents all spending that is non-urgent and is done largely for the purpose of curing (or potentially preventing) disease. Spending on a visit to a doctor, a test for blood cholesterol levels, and/or and spending on angioplasty are all examples of medical spending. This spending is non-urgent in the sense that there are no immediate consequences for an individual who chooses to forgo the spending; however, doing so may cause the individual's health to worsen.

Due to decreasing returns to scale within a period ($\psi < 1$), health is acquired most efficiently through small investments made each year. From the perspective of delayed care, ψ governs that intertemporal elasticity of substitution for healthcare spending — that is, how easily low spending today can be made up for by higher spending tomorrow. Consequently, the value of this parameter will play a key role in determining the extent to which reductions in delayed care lead to cost-savings.

Heath Emergencies: Each period, an individual faces an age- and health- dependent risk of experiencing a health emergency, denoted $\pi_x(h, a)$. An emergency carries two

⁵One may also wish to consider non-monetary investments into health (e.g. exercise). In Appendix Section B.1, I extend the model to allow for such inputs and show that this extended model is isomorphic to my baseline model under intuitive conditions.

consequences. First, it results in an additional one-time depreciation of an individual's health stock, described by δ^x in the health accumulation equation (3). Second, the individual must pay the costs of their emergency healthcare denoted x . These costs are stochastic and, conditional on experiencing an emergency, follow a log-normal distribution with a mean and variance that depend on the individual's age and health.

$$x(h, a) \sim \begin{cases} 0 & \text{with prob. } 1 - \pi_x(h, a) \\ \log N(\mu(h, a), \sigma(h, a)) & \text{with prob. } \pi_x(h, a) \end{cases} \quad (4)$$

Consistent with the literature (e.g. [De Nardi, French & Jones 2016](#)), the mean and variance of the shock are allowed to depend on an individual's health and age.

While health investment i includes all spending that could be deferred without immediate consequence, emergency expenditure represents urgent spending that cannot be delayed. A straightforward example would be bypass surgery administered at the ER to stop a heart attack or emergency surgery for the victim of a severe car crash. Although not technically compulsory, most patients aside from the few who leave the ER, ICU, or otherwise act Against Medical Advice (AMA) treat this spending as effectively compulsory; a doctor prescribes care, and the patient receives the treatment and later pays for it (or discharges the medical debt through bankruptcy).

With the health processes specified, it is apparent there are three reasons that individuals choose to invest in health. First, higher health reduces the likelihood of mortality which represents a direct increase in utility. Second, it reduces the probability of a health emergency which is costly for the individual. Finally, even conditional on having a health emergency, a healthy individual pays lower costs.

Imperfect Information: I allow for some fraction of individuals λ_χ to be poorly informed about the benefits and importance of health and health investment; this ends up being necessary for the model to match aggregate levels of health investment. In particular, I assume that a poorly informed individual with health status h perceives themselves as facing the mortality and emergency risks of an individual with health status $h^* > h$. When I bring the model to the data, I measure h using a health index falling in the interval $[0, 1]$ so a natural choice for h^* is

$$h^* = (1 - \chi)h + \chi \quad (5)$$

so that $\chi = 0$ reflects a perfectly informed individual and $\chi = 1$ reflects the extreme that an individual is completely ignorant of the benefits of health. An individual's information type is assigned at birth and remains unchanged for the entirety of their

life.

3.3. Health Insurance

Individuals can purchase health insurance to help pay for medical expenses and reduce the riskiness of emergency expenditure. Each period t , individuals choose to purchase (or not purchase) exactly one insurance plan from the set of plans for which they qualify. The individual is then covered by that plan in the next period $t + 1$. In this way, individuals may re-optimize their choice of plan each period but must commit to buying a plan before they realize their exact draw of stochastic shocks for the period of coverage.

Every insurance plan p is indexed by a tuple (λ, ν, d, P) representing the plan's copay rate λ , coinsurance rate ν , deductible d , and premium P . Emergency expenditure is covered through a standard deductible-coinsurance system; an individual facing emergency costs m in a single period must first pay up to their deductible d before any insurance coverage kicks in. Then the individual's insurance pays proportion $(1 - \nu)$ of any costs beyond the deductible within the period leaving the individual responsible for paying the remaining fraction ν .⁶ In total, the individual's share of emergency costs m is given by $\min(d, m) + \nu \max(m - d, 0)$.

The insurance plan also subsidizes non-emergency care through the copay rate λ . Operating similarly to the coinsurance rate, an individual must pay proportion λ of their medical expenditure i while insurance pays the remaining $(1 - \lambda)$. In addition to being a realistic feature of the model, such a subsidy makes sense from the perspective of the insurance company; the coverage of emergency expenditures introduces a moral hazard problem as individuals no longer face the full cost of their emergency expenditures and no longer fully internalize the benefits of investing in their health, and reducing their marginal cost mitigates this distortion.

Finally, the premium is the flat per-period cost of the individual's insurance plan. In theory, insurers would like to charge different prices to different individuals based on their assets, health, age, and other state variables. However, for tractability, I limit price discrimination to occur only via an exogenously-specified age profile $p(a)$, so that the insurer's problem involves only a single price P .

Altogether, an individual with insurance plan p , indexed by $(\lambda_p, \nu_p, d_p, P_p)$, who spends i on medical care and faces emergency expenditure m must pay out-of-pocket costs

⁶For simplicity, I abstract from out-of-pocket maximums, although these could be incorporated in the insurance scheme without much technical difficulty.

given by

$$\chi_p(i, m) = \underbrace{\lambda_p i}_{\text{Investment}} + \underbrace{\min(d, m) + \nu \max(m - d, 0)}_{\text{Emergency}} + \underbrace{Pp(a)}_{\text{Premium}} \quad (6)$$

Available Plans: There are four available health insurance plans — market insurance (available to all), employer-provided insurance (available only to some), Medicaid (available to sufficiently unproductive individuals), and Medicare (available to those older than 65) — as well as an option to be uninsured (which is treated symmetrically). The copay rate, coinsurance rate, and deductible (λ, ν, d) for each plan are taken to be exogenous and identical across individuals while the premium P is endogenous (for market-provided plans).

All individuals are eligible to purchase an individual plan from the market each period, denoted by $p = \text{IND}$. In addition, some individuals are eligible to purchase employer-provided insurance $p = \text{EMP}$. Although there is no *a priori* reason to prefer employer-provided insurance, when turning to the data, it is clear that employer-provided insurance plans offer lower deductibles and coinsurance/copay rates, on average, than market plans. Additionally, government subsidies will ensure that, despite better coverage, employer-based plans charge lower premiums than market plans.

Access to employer-provided insurance is not universal however. In reality, only individuals working for an employer who chooses to provide insurance or who previously worked for such an employer and remain covered through COBRA requirements have access to employer-provided programs. Replicating such a process in the model is difficult as the model lacks well-defined notions of job-switching or unemployment and tracking COBRA eligibility would involve many new state variables. Instead, I opt to model eligibility as a simple binary Markov process M .

In addition to the employer-provided and market plans, the government administers Medicare ($p = \text{MCR}$) and Medicaid ($p = \text{MCD}$). Medicare is available to all individuals age 65 or older. Although in reality there are many different coverage and plan decisions an individual must make *within* Medicare, I abstract from these and model Medicare as a single insurance plan. Medicaid, the US government public insurance for poor individuals, is made available to all individuals below a productivity threshold. Like with Medicare, I condense the complex reality of multiple Medicaid plans into a single representative plan. As part of this simplification, I model Medicaid availability as a function of productivity rather than earnings, simplifying away any labor market distortions.

Finally, individuals have the option to forgo insurance and enter the next period uninsured which is modeled as a fifth insurance plan with a deductible, copay, coinsur-

ance, and premium all equal to zero.⁷

Insurance Firms and Pricing: Employer and market insurance plans are each provided by their own representative insurance firm. These firms take the plan parameters as exogenous and face exogenous load parameters $\kappa_{\text{EMP}}, \kappa_{\text{IND}} > 1$ that summarize the overhead costs of administration — a firm that pays out x in total coverage must collect κx in premiums in order to break even for the period. Firms then set prices subject to a zero profit condition (i.e. perfect competition)⁸ which is given by

$$\int_{p=\text{EMP}} P_{\text{EMP}} p(a) d\Omega = (1 - s_{\text{EMP}}) \kappa_{\text{EMP}} \int_{p=\text{EMP}} \int (i^* + x - \chi_{\text{EMP}}(i^*, x) + P_{\text{EMP}} p(a)) df d\Omega \quad (7)$$

$$\int_{p=\text{IND}} P_{\text{IND}} p(a) d\Omega = \kappa_{\text{IND}} \int_{p=\text{IND}} \int (i^* + x - \chi_{\text{IND}}(i^*, x) + P_{\text{IND}} p(a)) df d\Omega \quad (8)$$

where Ω is the distribution of individuals across states and f is the (state-dependent) distribution of emergency expenditure shocks x . The LHS of each equation is the firm revenue given by the baseline premium chosen by the firm P_{EMP} and P_{IND} for individuals who purchases the plan last period. The RHS is the total outlays of the firm multiplied by the loading factor (note that P_{EMP} and P_{IND} are present here due to their inclusion in χ above). Finally, s_{EMP} is a proportional government subsidy for employer-based insurance.

3.4. Income and Labor Supply

Individuals younger than 65 years participate in the labor market and supply labor to a single sector earning wage w per efficiency unit of labor supplied.⁹ Individuals have a measure of labor productivity z which summarizes their total available efficiency units of labor. Labor productivity is given by the following stochastic process (with time

⁷While, generally speaking, the premium for the uninsurance “plan” could be positive, reflecting the presence of an individual mandate, the model will be estimated to match the pre-ACA United States which had no such requirement.

⁸The zero-profit assumption is of little consequence for this particular model. An alternative approach could use a constant loading factor $\hat{\kappa}$ and a constant markup σ . Because I directly observe the ratio of collected premiums to coverage outlays in individual expenditure data, these two parameters are not separately identified (i.e. I observe $\hat{\kappa}(1 + \sigma)$ directly); individuals do not care what proportion of their premiums accrue to operating costs ($\hat{\kappa}$) relative to profits (σ) and instead care only about this product.

⁹For simplicity, the baseline model makes no distinction between production of consumption or healthcare goods, essentially fixing the relative price between the two types of goods exogenously. Appendix Section B.3 lays out an extension of the baseline model with distinct sectors of production so that the relative price responds to policy in general equilibrium, and I report the robustness of my main quantitative results to this extension in Section 5.

subscripts suppressed where possible):

$$\begin{aligned}
z(z^p, z^s, a) &= e^{g(a)+z^p+z^s} \\
z_{t+1}^p &= z_t^p \\
z_{t+1}^s &= \rho z_t^s + \varepsilon_t \\
\varepsilon_t &\sim N(0, \sigma)
\end{aligned} \tag{9}$$

Here z^p is the individual's permanent productivity component which is invariant over their life-cycle while z^s represents a stochastic AR(1) component that leads to short-term fluctuations in income. Finally, $g(a)$ is a life-cycle component that depends on age a , allowing for deterministic life-cycle trends in productivity.

An individual with labor supply l earns pretax income given by

$$y_{\text{pre-tax}} = wz(z^p, z^s, a)l \tag{10}$$

Taxes and Retirement: Working individuals pay progressive income taxes that are used to fund government-provided health insurance (Medicare and Medicaid) and other expenditures. After-tax period earnings for an individual younger than 65 are given by

$$y_{a<65}(z^p, z^s, a, l_h, l_c) = T(wz(z^p, z^s, a)l)$$

where T is a monotonic (but unspecific for now) function.

At age 65, individuals retire exogenously and fix their labor supply l to 0 for the remaining periods of their life. Retired individuals receive social security income which depends on their permanent productivity and is given by the exogenous function $y_{a \geq 65}(z^p)$.¹⁰

3.5. Consumption, Savings, and the Budget Constraint

Individuals split their income between consumption c (the numeraire), medical expenditure i , emergency expenditure x , and assets b , which are risk-free and pay an interest rate of r each period. Markets are incomplete, and individuals cannot borrow (assets tomorrow b' must be weakly positive). The budget constraint of an age a indi-

¹⁰In reality, US social security income is determined by one's entire earnings history, but faithfully modeling this requires an intractable number of additional state variables. Following much of the literature, permanent income provides a good approximation of this average for everyone except the ex-post luckiest and unluckiest individuals who earned substantially more or less than their permanent income. Given the large increase in tractability for relatively little loss in accuracy, I opt for a simple model of retirement income.

vidual with assets b , insurance plan p , and productivities z^p, z^s is given by

$$\begin{aligned} c + b' + \chi_p(i, m) &= (1 + r)b + y_{a < 65}(z^p, z^s, a, l) & \text{if } a < 65 \\ c + b' + \chi_{\text{MCR}}(i, m) &= (1 + r)b + y_{a \geq 65}(z^p) & \text{if } a \geq 65 \end{aligned} \quad (11)$$

Should an individual experience a health emergency and incur costs beyond what they can afford without borrowing, the government will pay for some portion of their healthcare costs. In particular, the government will ensure that the individual achieves a consumption level of $\underline{c}$ and will pay exactly the amount of emergency expenditure necessary to ensure this level of consumption.¹¹ This is the closest model analogue to Medicaid's medically needy program.

3.6. The Individuals' Optimization Problem and Bellman Equation

There are eight individual-level state variables — assets b (1), health h (2), age a (3), permanent and temporary productivity (z^p, z^s) (4-5), and finally insurance plan, employer-provided insurance eligibility, and information status (p, e, χ) (6-8). For completeness, individuals also face the cross-sectional distribution of individuals across states Ω as an aggregate state variable; however, the analysis in Section 5 below will focus on stationary equilibria so that individuals need only to know steady-state prices.

The Bellman equation describing good-information individuals' optimization problem (when not bound by the consumption floor) is then written

$$\begin{aligned} V(b, h, a, z^p, z^s, p, e; \Omega) &= \max \bar{u} + u(c, l) + \\ &\quad + \beta(1 - \pi(h, a))\mathbb{E}[V(b', h', a + 1, z^p, z^{s'}, p', e'; \Omega')] \\ \text{s.t. } c + b' + \chi_p(i, m) &= (1 + r(\Omega))b + T(w(\Omega)z(z^p, z^s, a)l) \quad \text{if } a < 65 \\ c + b' + \chi_{\text{MCR}}(i, m) &= (1 + r(\Omega))b + y_{a \geq 65}(z^p) \quad \text{if } a \geq 65 \\ h' &= (1 - \delta_a - \delta_x)h + \phi i^\psi \\ p' &\in \{\text{EMP}, \text{IND}, \text{UN}, \text{MCD}\} \text{ according to eligibility} \\ \Omega' &= G(\Omega) \end{aligned} \quad (12)$$

where G is individuals' perception function which governs the expected evolution of the aggregate state. For compactness, I have suppressed the constraints describing the

¹¹To keep this policy sensible, I assume that individuals do not internalize the fact that this transfer responds to their budget and instead view it as a lump-sum. This prevents individuals from realizing that they face effectively zero marginal cost of health investment spending i (as any increase would be offset one-to-one with the transfers) and consuming extremely large amounts of care.

evolution of the stochastic processes (i.e. z^p, x, e), which can be found in the appropriate section above, as well as the non-negativity constraints for b' and i .

With the Bellman equation specified, the remaining components of recursive competitive equilibrium are standard, and the formal definition is relegated to Appendix B.2. The only detail worth noting is that the Bellman equation for poorly informed individuals is nearly identical to (12) with only the health-related stochastic processes (e.g. $\pi(h, a)$) substituted for their perceived counterparts ($\pi(h^*, a)$).

3.7. Delayed Medical Care

How does delayed care arise in the model? The key, as motivated in Section 2, is the discrete change in the individual's budget constraint that occurs at $a = 65$ and, in particular, that the individual now falls under Medicare ($p = \text{MCR}$), a zero-premium insurance plan that (as it will turn out) exhibits a relatively low copay rate λ_{MCR} , particularly when compared to uninsurance and marketplace insurance.

Like all consumption-savings models, individuals use their assets b to smooth their consumption. Within-period optimization between consumption and health spending dictates equality between the marginal benefit of each. The first-order condition is

$$u_c(c, l) = \frac{\phi\psi i^{\psi-1}}{p_h \lambda_p} \beta (1 - \pi(h, a)) \mathbb{E}\left(\frac{\partial}{\partial h'} V(h') | h\right) \quad (13)$$

where I have suppressed most of the inputs into the value function for brevity.

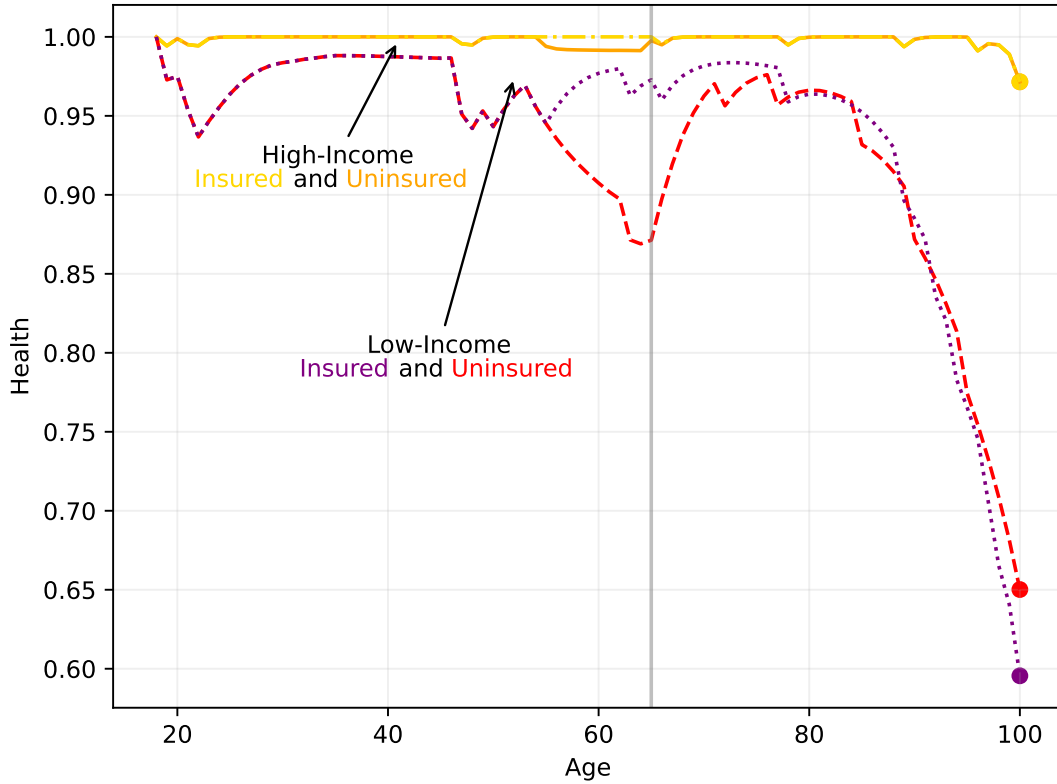
From equation 13, it is clear that as long as V exhibits diminishing marginal returns to health (as is the case in the estimated model), the marginal return to additional health $V_{h'}(h'^*)$ is inversely related to the individual's copay rate λ_p (as the individual acts to equate marginal benefits with marginal costs). Additionally, the envelope condition for the marginal value of health reveals that health is a forward-looking asset; part of the benefit of being healthy today is that one will continue to be healthy tomorrow.

Delayed care arises from the interaction of these two relationships. The individual expects to receive health insurance in the future, lowering λ_p , and thus the future marginal value of health. Because the value of health today depends on the value of health in the future, especially as the individual approaches the period they will receive insurance, this lowers the value of health today, and the individual reduces spending.

The economic intuition is straightforward. If health is going to be so much cheaper tomorrow (or in two or three periods) and a large portion of the value of health comes from its continuation value, why bother investing in health today? The most effective strategy is to treat health as an asset to be run down and then replenished once covered by insurance. This incentive is mitigated, however, by the decreasing returns to med-

ical care each period which ensures that medical care tomorrow is not too good of a substitute for medical care today, creating a link between the observed level of delayed care and the parameter ψ which I leverage in model estimation.

Figure 2: Life-cycle Health Dynamics for Four Individuals



Note: This figure displays life-cycle health dynamics for four individuals who receive an identical series of health and income shocks. The yellow line displays the health of an individual with high permanent income and access to employer-based insurance. The orange line displays the same for an individual with high permanent income and no health insurance. The purple and red lines display the health of low permanent income individuals with access to employer based insurance and no insurance respectively.

An Intuitive Example: Figure 2 provides an example of individual health investment behavior. The figure displays individual health (on the y-axis) over the lifecycle (the x-axis) for four singular individuals. They all receive an identical series of income and health shocks (with mortality shocks set to zero to prevent different years of death from precluding comparisons) but differ by their permanent income and their access to employer-provided health insurance. It is worth noting that in order to ensure that the cost of care and insurance remains a constraint, income for the low-income individuals is set just high enough to prevent them from qualifying for Medicaid, which would otherwise offset these effects.

The dynamics of health investment can be seen clearly in the figure. Early in life, the four individuals receive the same shocks but, because they spend more on health-care, the high-income individuals recover from shocks almost immediately and maintain high average health. Insurance status makes little difference in health dynamics at this point. Later in life, around age 58, the four individuals experience a series of bad health shocks just before retirement. Here the dynamics begin to diverge. As was the case with shocks early in life, the high-income individuals spend more on healthcare, recover from shocks faster, and keep their health at a higher level than the low-income individuals. However, as the individuals are approaching the age 65 threshold after which they will receive Medicare, they begin to delay care.

This delay of care is most noticeable in the differences between insured and uninsured low-income individuals. The insured individual (purple) continues to invest in health and mitigates the effects of the shocks to some extent. The situation is more dire for the uninsured individual (red) who spends almost nothing on healthcare and experiences substantial declines in health. After these individuals turn 65 and receive coverage through Medicare, their health slowly converges and the gap disappears around age 85 due to the uninsured individual's higher spending. Appendix Figure A.2 shows this directly by displaying annual healthcare spending for each of these two individuals over their life-cycle.

3.8. Production and Government Budget

With the household problem finished, closing the economy requires specify firm and government behavior, both of which are simple. There is a single representative firm producing output and operating in perfect competition with technology

$$Y = AL^\alpha K^{1-\alpha} \quad (14)$$

The government spends on social security, Medicaid (for the poor), Medicare (for the old), the subsidy for employer-provided insurance, and to ensure the consumption floor. The government also spends an exogenously-specified amount on a generic public good G that provides no utility. Following [Heathcote, Storesletten & Violante \(2017\)](#), the tax function T takes a simple monotonic form $\lambda_\tau y^{1-\tau}$. The government adjusts the level parameter λ_τ to ensure a balanced budget. In sum, the government budget con-

straint is given by

$$\begin{aligned}
& \underbrace{\int_{a < 65} (y_{\text{pre-tax}} - \lambda_{\tau}(y_{\text{pre-tax}})^{1-\tau}) d\Omega}_{\text{Revenues}} = G + \underbrace{\int_{p=\text{MCD}} (i^* + z - \chi_{\text{MCD}}(i^*, x)) df d\Omega}_{\text{Medicaid Outlays}} \quad (15) \\
& + \underbrace{\int_{a \geq 65} (i^* + z - \chi_{\text{MCR}}(i^*, x)) df d\Omega}_{\text{Medicare Outlays}} + s_{\text{EMP}} \underbrace{\int_{p=\text{EMP}} (i^* + z - \chi_{\text{EMP}}(i^*, x)) df d\Omega}_{\text{Emp. Provided Subsidies}} \\
& + \underbrace{\int_{a \geq 65} y_{a \geq 65}(z_p) d\Omega}_{\text{Social Security Outlays}} + \underline{c} \text{ Expenditure}
\end{aligned}$$

where the exact expression for expenditure on the consumption floor is suppressed due to being excessively cumbersome.

4. Data and Model Estimation

Bringing quantitative discipline to this model involves estimating many different parameters. Rather than provide an exhaustive description of how each parameter is estimated, this section summarizes the data and quasi-experimental results used for estimation, discusses the key parameters driving the main quantitative results, and provides intuition for how the data used in estimation discipline these parameters. A thorough discussion providing estimation details for every parameter can be found in Appendix Section C.

As mentioned in Section 2, the Medical Expenditure Panel Survey (MEPS) serves as the primary source of data for model quantification. Because the goal of the model is to understand the role that delayed care plays public health insurance expansion and because the approach to model estimation will rely on quasi-experimental moments from before the 2014 US expansion of Medicaid under the ACA, I use data from the 2013 wave of the MEPS. Thus the estimated model corresponds to the United States circa 2013.

Measuring Expenditure: The MEPS measures medical expenditure across a wide variety of categories. Bringing this data to the model requires a decision on where/how to make the distinction between spending on health investment (i) and spending on emergency health events (x). This is difficult to do even conceptually; an emergency bypass surgery also cleans arteries and results in a long-term improvement in health. I opt to categorize any spending that occurs in an emergency room or inpatient hospital setting as emergency spending and any other spending as investment, largely because this is unambiguously observed in the data. Appendix Table A.2 reports basic sum-

mary statistics for both investment and emergency spending. The overall patterns of spending are not surprising. Health emergencies are relatively rare with most individuals reporting no emergency expenditure in a given year regardless of age. Both types of spending are larger for older individuals and exhibit a strong right skew with means larger than medians.¹²

Measuring Individual Health: The MEPS also provides information on individual health status and outcomes. Mapping the complex reality of health to a simplified, abstract concept amenable to economic modeling is a classic difficulty in the macro-health literature. One solution has been to restrict health to a small number of discrete categories such as “Good” and “Bad” or ranging from “Excellent” to “Poor” (e.g. [De Nardi, French & Jones 2016](#), [Yogo 2016](#)). Particularly in the case of the latter, these categories are often self-reported subjective measures of health that may or may not be related to an individual’s actual health (see [Spitzer & Weber 2019](#), for an example).

To overcome these issues, I use the frailty index of [Hosseini, Kopecky & Zhao \(2021\)](#) which has the advantage of being largely objective and close-to-continuous, mapping cleanly to health h in the model. Frailty is constructed from a wide variety of yes or no questions about an individual’s health ranging from diagnoses (Have you ever been diagnosed with diabetes?) to cognitive limitations (Do you experience confusion or memory loss?) to common metrics known as Activities of Daily Living (Do you have difficulty getting dressed by yourself?), supplemented with some other objective measures such as an individual’s K6 score (a common measure of mental health). An individual’s frailty is determined by summing the number of “yesses” (referred to as the total number of health deficits) reported by the individual and normalizing by the total number of questions so that the index falls between 0 and 1.¹³

Subtracting the frailty index from 1 creates an analogous health index in which a value of 1 represents a maximally healthy individual who reports having no health deficits and 0 represents a minimally healthy individual who reports having every health deficit. I discuss the distribution of health (displayed in [Figure 4](#)) in more de-

¹²These patterns, as well as other results based the separation of spending into investment and emergent, are largely robust to alternative definitions such as limiting emergency expenditure to only that which occurs in an emergency room.

¹³[Hosseini, Kopecky & Zhao \(2021\)](#) discuss at length the usefulness of frailty as a measure of health and show that it is a strong predictor of many health outcomes, including medical expenditure and mortality, and that it outperforms self-reported measures of health. [Appendix Table C.3](#) replicates these findings and shows that the health index strongly predicts individual mortality, the probability of positive emergency expenditure, the total amount of emergency expenditure, and the variance of emergency expenditure (conditional on positive expenditure). The estimated coefficient for health is robust to the inclusion of a wide variety of controls including family income, race, sex, and geographic region, providing suggestive evidence that the index is not picking up variation in non-health-related latent variables.

tail in a later section on model validation. Here, it suffices to mention that it exhibits a strong left-skew with a high concentration of healthy individuals possessing health indices between 0.9 and 1.0 with a thin tail on the left-hand side. Very few individuals accumulate an extensive number of health deficiencies.

Measuring Outcomes: The functions determining health outcomes as a function of health h and age a ($\pi(h, a)$, $\pi_x(h, a)$, $\mu(h, a)$, and $\sigma(h, a)$ in Section 3 above) are important model objects as they directly determine the benefits of improved health. A substantial advantage of using the MEPS is that the panel structures allows these functions to be estimated directly; I use data from the survey round one year after an individual's initial survey to determine whether they experienced death and/or a health emergency during the year, as well as any emergency expenditures. I then estimate the probability of mortality, emergency expenditure and (conditional on positive expenditure) the mean and variance of emergency expenditure as a function of health and a quadratic in age; the results of these regressions are displayed in Appendix Table C.3. These estimated equations (particularly the versions without other controls in columns 1, 4, 7, and 10) serve as the functions $\pi(h, a)$, $\pi_x(h, a)$, $\mu(h, a)$, and $\sigma(h, a)$ in the quantitative model.

4.1. Key Parameters and Model Estimation

An exhaustive list of model parameters and their estimated values can be found in Appendix Section C (Appendix Tables C.1, C.2, C.4, and C.5 in particular). Here I discuss the key parameters driving the model results and their quantification.

Preferences: Period utility takes the King-Plosser-Rebelo (KPR) functional form given by

$$u(c, l) = \bar{u} + \frac{1}{1 - \sigma} c^{1 - \sigma} (1 - \kappa(1 - \sigma) l^{1 + \frac{1}{\nu}})^{\sigma} \quad (16)$$

with the Frisch elasticity set to unity.

As discussed previously, the parameter $\bar{u}$ governs individuals' willingness to trade off between consumption and mortality and is chosen to match an average Value of Statistical Life (VSL) of 10 million USD (Viscusi & Aldy 2003) — a very typical value used across many contexts.

It is also important to mention the value of σ , the coefficient of relative risk aversion, which is set at 2. This value is commonly used in both the consumption-savings and taxation literatures. However, Murphy & Topel (2006) and Hall & Jones (2007) demonstrate that this parameter plays a critical role in determining how an agent's willingness to trade-off consumption and mortality changes with income. Intuitively, this occurs because the intertemporal elasticity of substitution governs how much the agent values living to see future consumption. If the IES is high, consumption today is a good sub-

stitute for consumption tomorrow, and the agent can consume heavily today and not worry about whether or not they will be alive to see tomorrow’s consumption. Thus the CRRA/IES parameter is doing “triple duty”; it governs the agent’s risk tolerance (as in the consumption-savings literature), their labor supply response to income shocks (as in the taxation literature), and the income elasticity of their VSL (as in the sparse macro-health literature). The chosen value of 2 gets all three behaviors roughly correct.¹⁴

Insurance Plans: Conditional on the preference parameters and the distribution of health shocks, willingness to purchase health insurance depends on the parameters of insurance plans — namely the coverage provided and the loading factors dictating premiums. For each plan type (Employer, Marketplace, Medicaid), I compute parameters for a representative plan by computing the portion of total of spending in a category that comes out-of-pocket. For example, the copayment rate for employer-provided insurance is estimated by dividing the total amount of out-of-pocket non-emergency spending by the total amount of non-emergency spending among those with employer-provided insurance. Coinsurance rates and rates for marketplace and Medicaid insurance plans are estimated similarly (with the caveat that coinsurance rates must account for deductibles). The parameters for the Medicare insurance plan are set to their statutory values.¹⁵ Loading factors κ_{EMP} and κ_{IND} are estimated similarly by dividing total premiums collected by total outlays for by employer-provided and marketplace-purchased plans.

Appendix Table A.3 displays the estimated values for all insurance plan parameters. The estimated copay rates are 20 percent and 25 percent for employer-provided and marketplace plans respectively, meaning that covered individuals pay for 20 or 25 percent of their non-emergency medical spending. Coinsurance rates are lower at 6 and 13 percent. The estimated loading factor for the marketplace plan is 1.43 compared to 0.98 for the employer-provided plan (employer-provided plans pay out slightly more in benefits than they collect in premiums, at least from households). I interpret the difference in loading factors to be result of government subsidies rather than structural advantages in administration for employer-provided plans. In other words, subsidies cover the remaining 45 percent of payments that would need to be collecting in premiums. These results are in line with typical intuition; employer-provided plans have

¹⁴As an alternative approach, one could use a variant of Epstein-Zin preferences, such as those suggested in Córdoba & Ripoll (2017), to adjust parameters governing each behavior separately, but this adds substantial computational complexity for little gain in accuracy.

¹⁵Fixing parameters to their statutory rates is possible for Medicare but not Medicaid due to the fact that the latter exhibits substantial variation in coverage from state to state. Thus estimating a representative Medicaid plan is dramatically simpler than attempting to somehow aggregate the various state-level plans.

better coverage and do so with lower premiums.

The Productivity of Health Spending: Intuitively, the health investment productivity parameter ϕ largely dictates the link between delayed care and mortality. A low value corresponds to a world in which spending on healthcare does not lead to very substantial gains in health, and shocks end up being a larger determinant of health than spending. As a consequence, delaying healthcare spending has little impact on mortality, and reductions in delayed care do not lead to substantial reductions in mortality. Conversely, a high value of ϕ corresponds to a world where health is primarily determined by health investment, and delaying care has large effects on mortality.

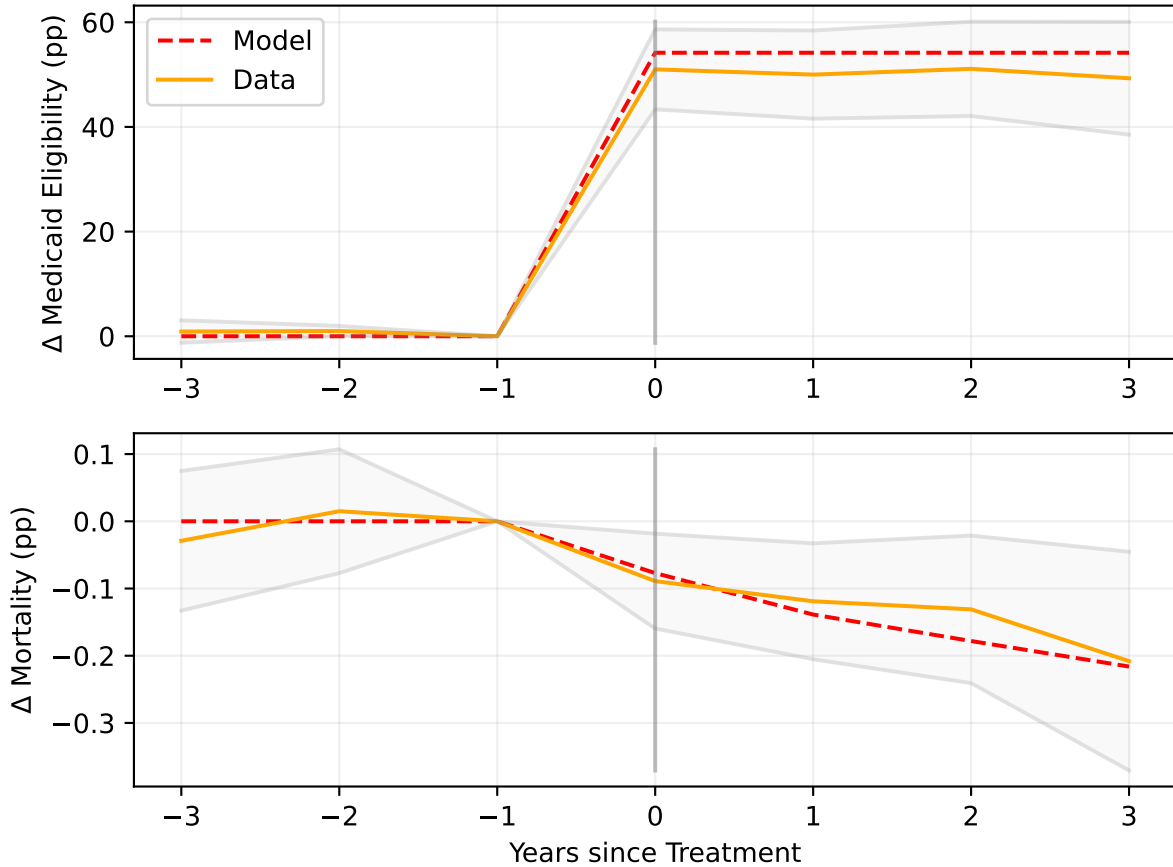
I leverage this relationship to discipline ϕ by estimating the model to match a quasi-experiment that estimated the impact of public health insurance (Medicaid) provision on mortality, lending credibility to the quantitative predictions of the model. [Miller et al. \(2021\)](#) use the fact that the 2014 Affordable Care Act (ACA) increased the income threshold in Medicaid eligibility only in certain states. Combined with administrative social security data on mortality, they use a diff-in-diff framework comparing states that expanded to those that did not expand and estimate that Medicaid expansion reduced mortality for newly eligible individuals between 55 and 64 years old about 0.14 percentage points on average in the first four years after expansion.

Estimating the model to match this moment requires producing a model analogue for the diff-in-diff estimation strategy. I begin with two Medicaid eligibility thresholds, $\underline{z}_{\text{PRE}}$ which corresponds to the threshold in 2013 (the year the model is estimated to match), and $\underline{z}_{\text{POST}}$, which corresponds to the threshold after expansion in 2014. I take the steady-state distribution of individuals from the 2013 steady-state and replicate the sample selection procedure of [Miller et al. \(2021\)](#) by restricting the sample to include only those who are between 55 and 64 years old and who would qualify for Medicaid after expansion (i.e. have $z < \underline{z}_{\text{POST}}$). I then consider two different worlds, one in which the eligibility cutoff remains at $\underline{z}_{\text{PRE}}$ (control) and the other in which it is increased to $\underline{z}_{\text{POST}}$ (treatment), and simulate the sample forward in each scenario. The model equivalent of the diff-in-diff estimator is constructed by taking the simple difference between mean outcomes in the treatment and control worlds as the “pre-period” outcomes are identical between treatment and control by construction.

The final detail is choosing appropriate values the eligibility cutoffs. The value for $\underline{z}_{\text{POST}}$ is straightforward; states that expanded Medicaid under the ACA were required to offer coverage to anyone earning less than 138 percent of the Federal Poverty Level (FPL), so I choose $\underline{z}_{\text{POST}}$ so that the model replicates this fact. The choice of $\underline{z}_{\text{PRE}}$ is more complicated as eligibility cutoffs varied dramatically across states before the ACA. In

order to maximize comparability between model and experimental evidence, I choose $\bar{z}_{\text{PRE}}$ to match the “first-stage” of the experiment, namely, the increase in Medicaid eligibility experience by the treated group (roughly 50 percentage points).

Figure 3: Effect of Medicaid on Mortality: Model vs Data



Note: This figure displays the impact of Medicaid expansion on Medicaid qualification and mortality of low-income 55-64 year old adults. The solid orange line displays the effects estimated in [Miller et al. \(2021\)](#) using a diff-in-diff design. The grey bands represent the estimated 95% CI. The dotted red line displays the change in mortality in the calibrated model.

Figure 3 displays the results of the diff-in-diff estimation in both the data (orange solid line with 95 percent confidence intervals shown in gray) and the model (red dotted line). The top panel displays the increase in Medicaid eligibility. The bottom displays the estimated decline in mortality for the first three periods after expansion (the longest horizon reported in [Miller et al. 2021](#)). The model closely matches the increasing in coverage and the overall decline in mortality (unsurprising, as these are targeted moments). Notably, it also closely matches the dynamics of the decline in mortality, which starts small and then increases slowly up to a factor of about 3x in the fourth year after

expansion, despite this not being targeted. This growing decline is intuitive as every year that passes is an additional year that treated individuals will build health relative to control individuals and also highlights one advantage of model-based analysis — the model can be used to analyze the full long-run effect expansion which may be much larger than the impact estimated in the first few years.

As a brief aside, one could be concerned that the estimate from [Miller et al. \(2021\)](#) is at odds with the broader literature linking insurance and mortality. Appendix Figure A.6 plots the point estimates for the impact of health insurance coverage on all-cause mortality from this study, as well as four other studies from the literature (including the evaluation of the Oregon Health Insurance Experiment from [Finkelstein et al. 2012](#)). The figure makes it clear that although some other studies have found statistically insignificant impacts, the point estimate from [Miller et al. \(2021\)](#) is actually among the smallest of those in the literature, suggesting that the difference in conclusions between studies is due to differences in power, rather than ambiguity over the impact of health insurance.

The Within-Period Returns to Scale of Health Spending: While ϕ is tightly linked to mortality, the returns-to-scale parameter ψ largely determines the magnitude of cost-savings. This parameter acts as the intertemporal elasticity of substitution for health spending. A low value indicates that low health spending today cannot be easily offset by higher health spending tomorrow and that achieving the same level of health through low spending in one period and higher spending in the next requires greater total spending. Consequently, a low value for ψ indicates that reductions in delayed care due to public insurance expansion can potentially lead to large reductions in aggregate healthcare expenditure.

The main source of quantitative discipline for ψ comes from the results in Section 2 above that estimate the jump in healthcare utilization that occurs at age 65 and, consequently, the extent to which individuals delay care. The relationship follows a revealed preference logic. If the value of ψ is low, then delaying care is very costly in terms of lost health and, consequently, individuals will not choose to delay. Thus the measured jump will be small. Conversely, a high value of ψ means that the delaying care is not very costly, and low spending today can be made up for more easily by high spending tomorrow. In response, individuals will choose high levels of delayed care. This relationship demonstrates an unintuitive “paradox” — the more delayed care that is observed, the lower the (implied) cost of any given quantity of delayed care, making it hard to evaluate its aggregate consequences without a quantitative model.

Of the multiple measures of utilization used to estimate the jump in Table 1, I opt to

target increase in utilization as measured by the increase in expenditure (i.e. via actual amounts paid to providers). The estimated jump in expenditure on Office-Based Visits is 2.2 percent with null estimates for other categories. Office-Based Visits make up 42 percent of total non-emergency expenditure, so the jump in total expenditure targeted in model estimation is 0.9 percent ($= 0.022 * 0.42$).

The model period is one year, making it impossible to replicate the month-by-month approach employed in Section 2 in the model. One option is to simply construct the model-equivalent moment as the increase in expenditure from age 64 to 65, but the upward trend in expenditure over the lifecycle means that this moment would overstate the jump due to Medicare. Instead, I opt to use the amount by which the increase in expenditure from 64 to 65 exceeds the increase from 63 to 64. Subtracting away the pre-Medicare upward trend in this way isolates the increase in expenditure that occurs due to coverage, closely corresponding to the idea of the jump measured in Section 2.

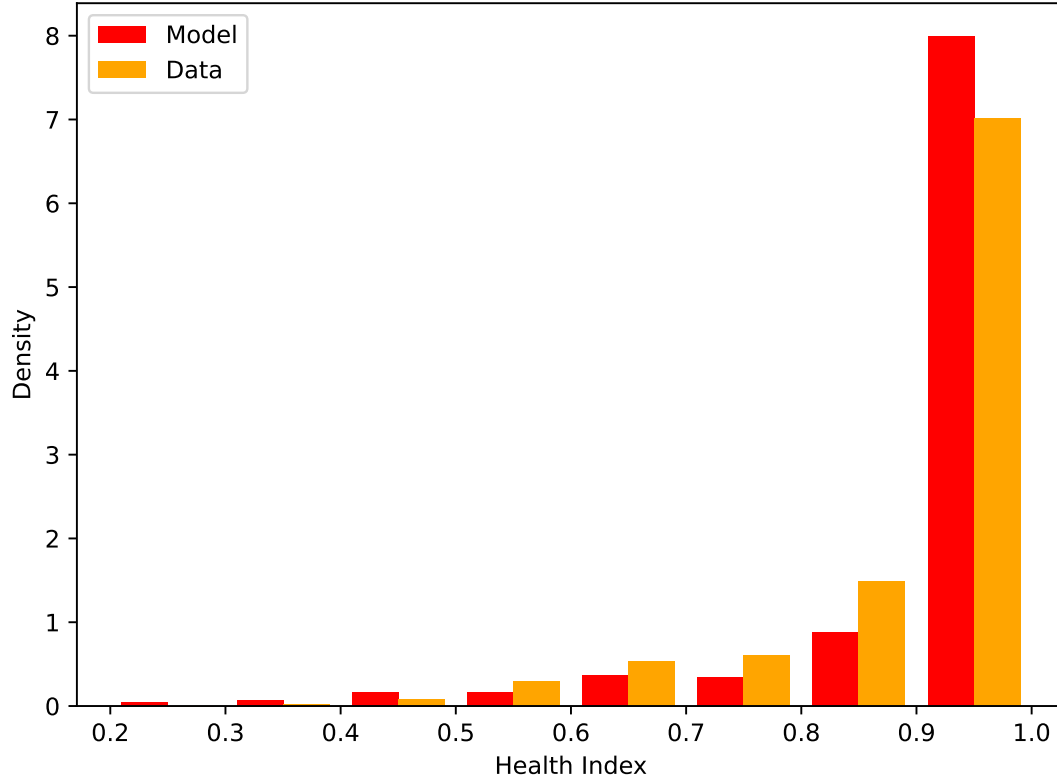
4.2. Discussion of Parameter Values and Model Validation

As a first pass at model validation, Figure 4 compares the distribution of health in the estimated model to that observed in the MEPS using a pair of overlapping histograms with the red histogram corresponding to health in the model and the orange histogram corresponding to the MEPS. Although the mean of this distribution is the only moment targeted in model estimation, it is clear that the model successfully replicates the stark features of the data, including bunching at the top and a thin left tail. The model slightly overpredicts the variance of the distribution as there are more individuals with extremely low health — $h \in (0.2, 0.5)$ — and in perfect health in the model than are observed in data. The model also successfully matches the fact that health is highly persistent; the estimated coefficient of an AR(1) process in health is 0.996 in the model, compared to 0.991 estimated from data in Hosseini et al. (2021). Given the natural and (relatively) parsimonious description of the health accumulation process, I view this as a substantial success of the model.

It is also worth checking if the model produces reasonable values for various aggregate statistics that are relevant to public health insurance coverage and costs. Most importantly, the model is able to deliver reasonably accurate predictions for the cost of public health insurance expansion under the 2014 ACA. The estimated model implies that such an expansion requires an annual increase of \$50 billion in Medicaid expenditure. In reality, expansion resulted in an increase in annual expenditure of expenditure \$69 billion by 2017. (CMS 2018)

The model also closely matches Medicaid enrollment rates. In the pre-expansion steady-state of the model, 8.4 percent of individuals qualify for and enroll in Medicaid.

Figure 4: The Distribution of Health in the Data and Model



Note: This figure displays the distribution of health in the Medical Expenditure Panel Survey, as measured by the health index based on [Hosseini et al. \(2021\)](#), in orange. The distribution of health in the calibrated model is displayed in red. See the discussion in Section 4.2 for more details.

In 2013 (the year before expansion), [CMS \(2014\)](#) reports that the analogous rate in data is between 6.1 and 10.2 percent.¹⁶ In the post-expansion steady-state, 18.9 percent of individuals are enrolled in Medicare. Although not directly comparable to the data as not all states expanded insurance access, a back-of-the-envelope adjustment based on the fact that roughly two-thirds of the population live in states that expanded access suggests that 15.4 percent ($= \frac{1}{3}8.4 + \frac{2}{3}18.9$) is a reasonable value to compare to data. The observed enrollment rate in 2018 was between 11.0 and 15.2 percent ([CMS 2018](#)). So the model closely matches but slightly overstates the extend of enrollment increase, at least using this metric.¹⁷

¹⁶The ambiguity in this range stems from the fact that some adults qualify for Medicaid through disability requirements rather than income requirements. While conceptually the closest analogue to the model would only count adults who qualify through income requirements, I do not observe whether or not a disabled individual would qualify through income requirements were they not disabled. Given the large income penalties associated with disability, it is not hard to imagine that there are many such individuals, making it impossible to pin down the rate more precisely.

¹⁷It is worth noting, however, that Medicaid enrollments have continued to rise over time, exceeding 20

One area where the model is less successful is the relationship between individual health and age. Though the model accurately reproduces the life-cycle pattern of health spending (Appendix Figure A.3), it generates a covariance between health and age of -0.5 while the corresponding value in the data is -0.9. Appendix Figure A.4 plots health as a function of age in both the data and the model, revealing that this discrepancy is largely due to the model over-predicting late-in-life health. This could be a substantial issue if it leads the model to over-predict lifespans, under-predict mortality, and wrongly predict how the two vary with income (which determines public insurance coverage) as these are important determinants of the magnitude of the mortality channel.

Fortunately, the model performs well on these metrics. It produces an average life expectancy at age 18 of slightly over 80 years; the equivalent number (assuming a population of half males and half females) from the US Social Security Administration Life Tables is just over 79 (SSA 2023). Figure 7 in Section 5 below also shows that the model accurately replicates life-cycle mortality, as measured either directly in the MEPS or via the SSA Life Tables. Importantly, the model also does reasonably well in reproducing cross-sectional variation in life expectancy with respect to income. In the model, those in the 75th percentile of the income distribution live 3.9 years longer than those at the 25th percentile. The equivalent number from Chetty et al. (2016) is 5.5 years (~ 7 years for men and ~ 4 years for women).

Values of Health Parameters: The productivity and returns to scale parameters of the health production function (ϕ and ψ) are central to the model’s quantitative predictions, so it is worth briefly discussing them. The estimated value of ϕ is about 0.11 and the returns to scale parameter is estimated to be 0.22. As an intuitive example, these estimated values imply that an otherwise perfectly healthy individual without any other health shocks must spend roughly 550 USD (combined out-of-pocket and covered) in order to erase one health deficit.¹⁸ Further, the estimated value of ψ implies that the marginal effectiveness of health spending within a period declines very rapidly, making it very slow and expensive to cure an accumulation of deficits. For an individual with the average level of annual health spending (2,600 USD) each year, curing an additional deficit requires doubling their annual spending for slightly over four years. While there is no direct data analogue to compare these values against, they seem at least reasonable and roughly in line with intuition.

percent of US adults by 2025 (KFF 2025). Though at this time horizon it is difficult to distinguish between the long-run impacts of expansion versus other trends, this at least suggests that the model overstating the observed enrollment increase may simply be a product of the measured time horizon being too short.

¹⁸One deficit = 0.042 units of h

5. Quantitative Results

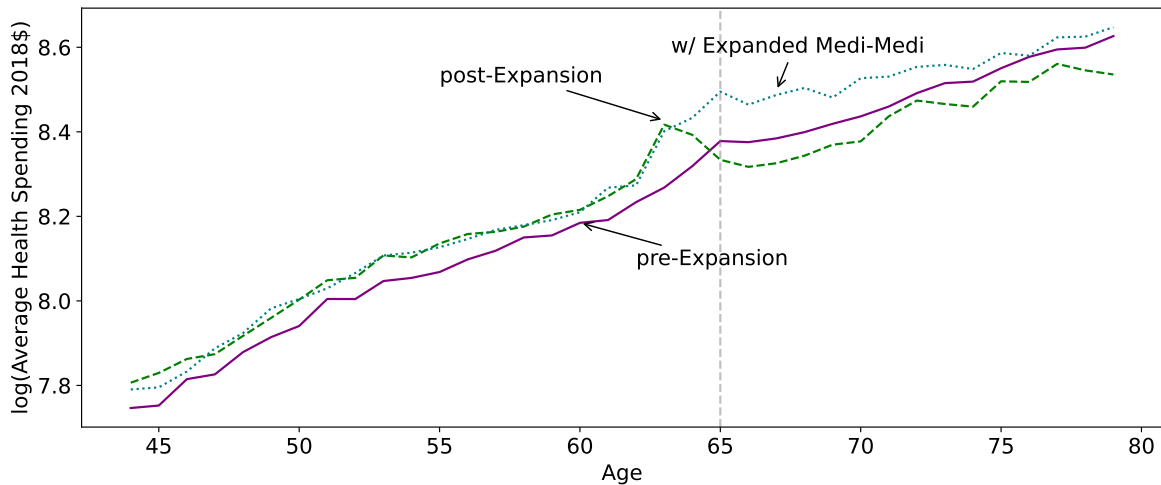
I use the estimated model to evaluate the results of an expansion in public health insurance similar to the 2014 ACA Medicaid expansion. Starting from the model steady-state under a Medicaid productivity cutoff of $\bar{z}_{\text{PRE}}$, I increase the cutoff to $\bar{z}_{\text{POST}}$ and simulate the economy forward. The values for $\bar{z}_{\text{PRE}}$ and $\bar{z}_{\text{POST}}$ are taken directly from the model estimation exercise of the previous section. In order to isolate the mechanics of delayed care, I focus largely on comparisons in partial equilibrium absent changes in prices, taxes, or premiums. I do, however, briefly discuss general equilibrium in order to assess the robustness of the main results and perform welfare calculations.

Reduction in Delayed Care: A natural first question is the extent to which expansion actually impacts the use of care around the age 65 threshold. Figure 5 displays the log of average healthcare spending for individuals aged 45 to 80. The purple line displays the mean for the pre-expansion steady state while the dashed green line displays the mean for the post-expansion steady state. The figure reveals that expansion eliminates delayed care and, surprisingly, even reverses this phenomenon; post-expansion spending peaks at age 63 and then falls 9 percent at age 65 relative to this peak. This occurs because of the fact that Medicaid's copayment rate of 2 percent is more generous than Medicare's 20 percent and expansion under the ACA only extends to non-elderly adults (those younger than 65). Thus in addition to reducing the number of insured individuals who are incentivized to delay care, the expansion also creates some individuals whose cost of healthcare will increase substantially at 65 and are thus incentivized to stockpile health while approaching this threshold.

In reality, this new incentive was dampened by the fact that many (though not all) states responded to expansion by also increasing elderly Medicaid coverage rules in order to come closer to matching the new eligibility rules for the non-elderly, substantially increasing the number of people with "Medi-Medi" coverage for whom Medicare pays the majority of the cost of care and Medicaid pays any difference between the higher Medicare rate and the lower Medicaid rate. Though a detailed modeling of Medi-Medi (the particulars of which are often quite involved and idiosyncratic at the state level) is unfeasible in this paper, it is worth extending the baseline model with a stylized version in order to check that this is not entirely driving the results. I do this by adding a Medi-Medi plan for individuals 65 or older with parameters that are identical to the Medicaid plan for the young¹⁹ and a simple qualification rule: if you qualified for Medicaid at age 64, you qualify for Medi-Medi for the rest of your life. The results of this

¹⁹The individual does not care what portion of their healthcare costs are paid by Medicare vs Medicaid and only cares about the final marginal cost of care.

Figure 5: Model Implied Delayed Care Pre- and Post- Expansion



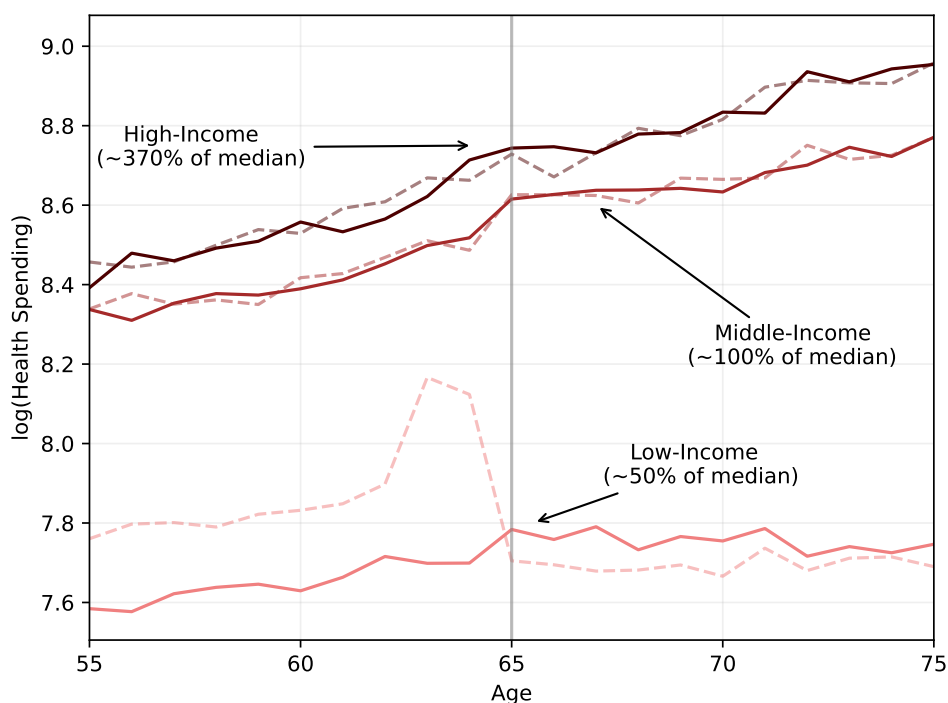
Note: This figure displays the log of average health spending. The purple line shows spending from the pre-expansion steady-state of the calibrated model while the dotted green line shows spending from the post-expansion steady-state. The jump at age 65 is estimated using a regression discontinuity design with a cutoff at age 65.

extension are displayed in Figure 5 via the dotted blue line and assuage the potential concerns. The increase in spending due to expansion while approaching the age 65 threshold is similar, if not higher, under the extension.

Turning to heterogeneity, Figure 6 displays average health spending before (solid lines) and after (dashed lines) expansion for three different income bins. The patterns revealed by the figure are unsurprising. The increase in care below the age 65 threshold that occurs with expansion is driven entirely by the lowest income bin (who have average income equal to roughly 50 percent of the median). These are the individuals who are both likely to be uninsured before expansion and gain insurance as a result of expansion. In contrast, individuals in the middle income bin (average income roughly equal to the median) are still fairly likely to be uninsured and consequently exhibit a jump in consumption at age 65; however, they remain unlikely to qualify for Medicaid even after expansion. Thus expansion does little to mitigate their delay, and the jump remains. Finally, the highest income individuals experience very little delay and are essentially unaffected by expansion.

Reduction in Mortality: How substantial is the reduction in mortality due to less delayed care? Figure 7 displays mortality as a function of age in the pre-expansion (solid purple line) and post-expansion steady states (dashed green line) along with the data values measured in the MEPS (orange dots) and as projected by the SSA Life Tables (dotted red line). The decline in mortality is substantial — recall that the model is esti-

Figure 6: Pre-Expansion Health Spending by Age and Income

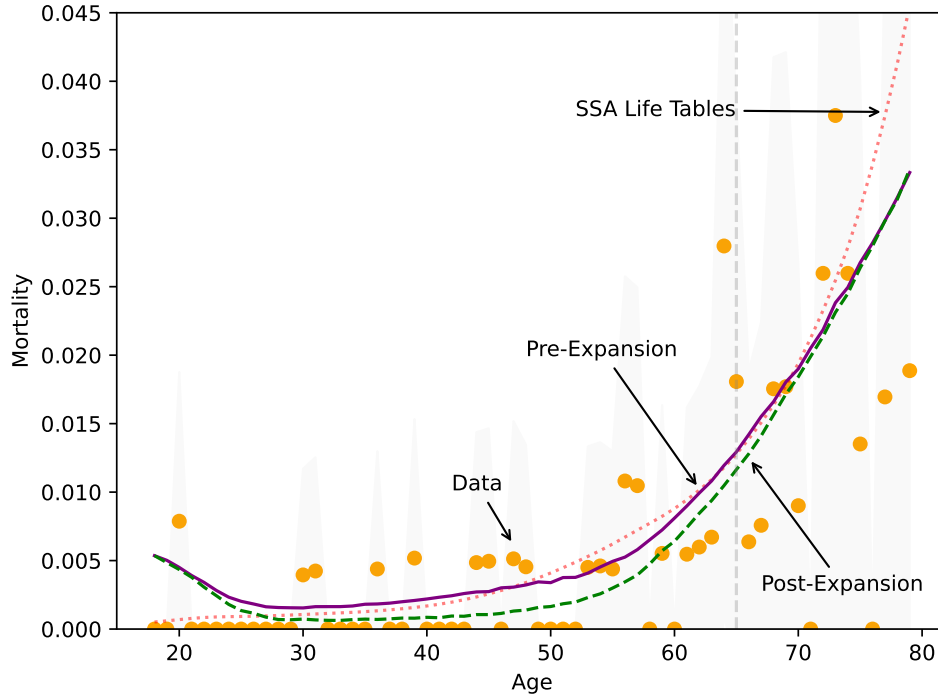


Note: This figure displays the log of average health spending as a function of age and permanent income in the pre-expansion steady-state of the model.

mated to match the decline in mortality measured in [Miller et al. \(2021\)](#) — and persists even after age 65, consistent with the notion that expansion and the reduction in delayed care lead to long-run improvements in health for impacted individuals. Looking from ages 60 to 70, average annual mortality declines by about 0.13 percentage points resulting in a 1.2 percentage point higher likelihood of survival to age 70 conditional on reaching age 60.

This reduction in mortality all-but-fades by approximately age 75 and even slightly increases near age 80 and beyond. While some of this is certainly a result of the gap between pre-expansion and post-expansion individuals closing, as everyone older than 65 has the same coverage in both cases, it primarily reflects the selection effect that occurs due to the fact that individuals who experience bad series of health shocks are surviving longer in the post-expansion case. Reflecting this, average health of those older than 75 counter-intuitively drops by about 0.01 (one-quarter of a deficit) after the expansion. While this may appear malignant at first glance, it is actually the result of improved longevity and represents an increase in welfare.

Figure 7: Model Implied Mortality Pre- and Post- Expansion



Note: The purple line shows mortality from the pre-expansion steady-state of the calibrated model while the dashed green line shows mortality the from post-expansion steady-state. The orange dots display mortality from the 2018 MEPS data (95 percent CI shaded), and the dotted red line displays mortality rates from the SSA Life Tables.

5.1. Delayed Care and the Cost of Expansion

The reduction of delayed care due to public health insurance expansion impacts the cost of expansion through both the cost-effectiveness channel (treating disease earlier is more cost effective) and the mortality channel (treating disease early saves lives and increases required Medicare outlays).

Table 3 displays the steady-state per-period cost of expansion in the model and the resulting change in Medicare expenditures, as well as the results of some counterfactuals aimed at distinguishing between the various model channels. Column (1) corresponds to the baseline expansion counterfactual which, recall, occurs in partial equilibrium. Medicaid coverage increases by 9 percent, requiring an increase in Medicaid (public insurance for the poor) spending of about 0.3 percent of GDP. More interestingly, Medicare (public insurance for the old) expenditures decrease by 0.02 percent of GDP, indicating that the cost-effectiveness channel quantitatively dominates the mortality channel. Consequently, the true cost of expanding Medicaid is lower than simply looking at the change in Medicaid outlays would imply. As a ratio, Medicare costs de-

cline by approximately \$8 for every \$100 spent on expanding Medicaid, reducing the true cost of expansion by 8 percent relative to the apparent cost.

Table 3: Changes in Healthcare Spending as a Result of Expansion

Variable	(1) Baseline	(2) Exogenous $\tilde{\pi}$	(3) in GE
Medicaid Coverage (% of Pop.)	+10%	+10%	+10%
Medicaid Spending (% of GDP)	+0.31%	+0.42%	+0.30%
Medicare Spending (% of GDP)	-0.02%	-0.13%	-0.03%
Medicare Savings per \$100	\$8	\$31	\$12

Note: This table displays the increase in Medicaid coverage (in percentage of the population), increases in Medicaid and Medicare spending (in percentages of GDP), and savings on Medicare per \$100 spent on Medicaid (in USD) implied by the estimated model. Column (1) displays these changes for the baseline model while Column (2) displays these changes in an auxiliary model with exogenous mortality (explained in detail in the text) and Column (3) displays these changes in general equilibrium when prices and taxes are allowed to adjust.

This modest reduction in Medicare outlays, however, does not distinguish between the case where both the cost-effectiveness and mortality channels are themselves small and the case where the two channels are quantitatively large but move in opposite directions, resulting in a small net impact. To examine the two channels in isolation, I introduce an auxiliary model in which the mortality function $\pi(a, h)$ is replaced by an alternative function $\tilde{\pi}(a)$ that does not depend on health. This mechanically eliminates the mortality channel as improvements in health no longer reduce mortality, leaving only the cost-effectiveness channel. However, it also eliminates the primary incentive for individuals to invest in their health. In order to account for this and, additionally, minimize any differences between the two counterfactuals not arising from the mechanical fixing of mortality rates, I prevent individuals from re-optimizing their behavior in response, and simply assign their health spending according to their previous policy functions. I choose $\tilde{\pi}$ so that average mortality for each age a in the auxiliary model is identical to average (pre-expansion) mortality in the baseline model.

Column (2) display the impact of expansion in this counterfactual with exogenous mortality. Expansion is slightly more costly (as removing the relationship between health and mortality leads some low-health, high-utilization individuals to survive longer), but the largest change is in the reduction in Medicare outlays which increase in magnitude by a factor of six. As a result, the decline in Medicare costs per \$100 dollars

spent on expansion increases to \$31, and the true cost of expansion in this counterfactual is much lower than the apparent cost would suggest.

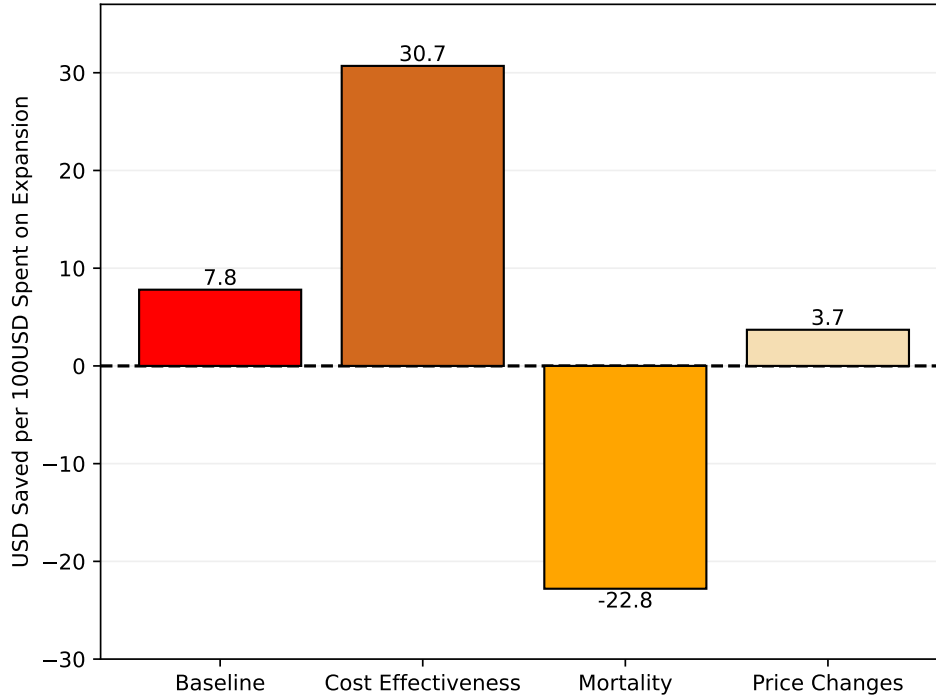
Because this counterfactual removes the mortality channel while keeping the cost-effectiveness channel intact, I interpret this figure of \$31 saved on Medicare per \$100 spent on Medicaid to be the quantitative impact of cost-effectiveness and \$23 saved per \$100 spent (\$31 less \$8) to be the impact of the mortality channel. Both these figures are, in isolation, substantial, indicating that both channels are quantitatively important. Similarly, the conclusion of the baseline model in Column (1) that the cost-savings of expansion are modest is the result of two important channels pushing against each other leading to a small net effect, rather than arising from the fact that the channels themselves are inconsequential.

Column (3) displays the results of expansion in an extension of the baseline model with the goal of confirming that the results reported on column (1) are robust in general equilibrium. In particular, the extended model allows for separate consumption and healthcare sectors (so that expansion impacts the relative price of healthcare), adjustment in insurance premiums in response to changing healthcare prices, and adjustments in taxes. The details are summarized in Appendix Section B.3. In general equilibrium, the results are similar to the partial equilibrium results of the baseline model. Medicaid spending increases by slightly less and Medicare spending decreases by slightly more, both as a result of the fact that the price of healthcare increases slightly (by about 0.5 percent). This impact is larger for Medicare, as the elderly spend a larger portion of their income on healthcare, and, as a result, the Medicare savings per \$100 spent on expansion increase slightly to \$12.

For expositional clarity, Figure 8 displays the total decline in Medicare expenditures per \$100 spent on expansion as well as the contributions of each component. In total, expenditures decrease by about \$8. This is the net result of a substantial decline of \$31 due to the cost-effectiveness channel and an offsetting increase of \$23 due to the mortality channel. In the general equilibrium extension, changes in prices and taxes also have a small impact, contributing about \$4 per \$100. Overall, these results suggest that the long-run cost savings from expanding public health insurance are modest, stemming from the fact that the increased costs due to the resulting decline in mortality serve to offset about three-quarters of the potential savings from cost-effectiveness.²⁰

²⁰One may naturally wonder to what extent this conclusion is the result of the estimated parameter values, and thus the moments targeted in estimation, versus simply being baked into the structure of the model or, similarly, how sensitive the main conclusions are to key parameter values. Appendix Table A.4 performs sensitivity analysis by examining how these conclusion changes when key model parameters are changed. Overall, while this conclusion is robust to reasonable variation in many single parameters, making large changes to multiple key parameters can increase the size of the mortality channel to the

Figure 8: Medicare Savings per \$100 Spent on Expansion



Note: This figure displays the total savings on Medicare per \$100 spent on Medicaid expansion, as well as the contributions of the cost effectiveness, mortality, and price change channels. Refer to the text for details on how this decomposition is constructed.

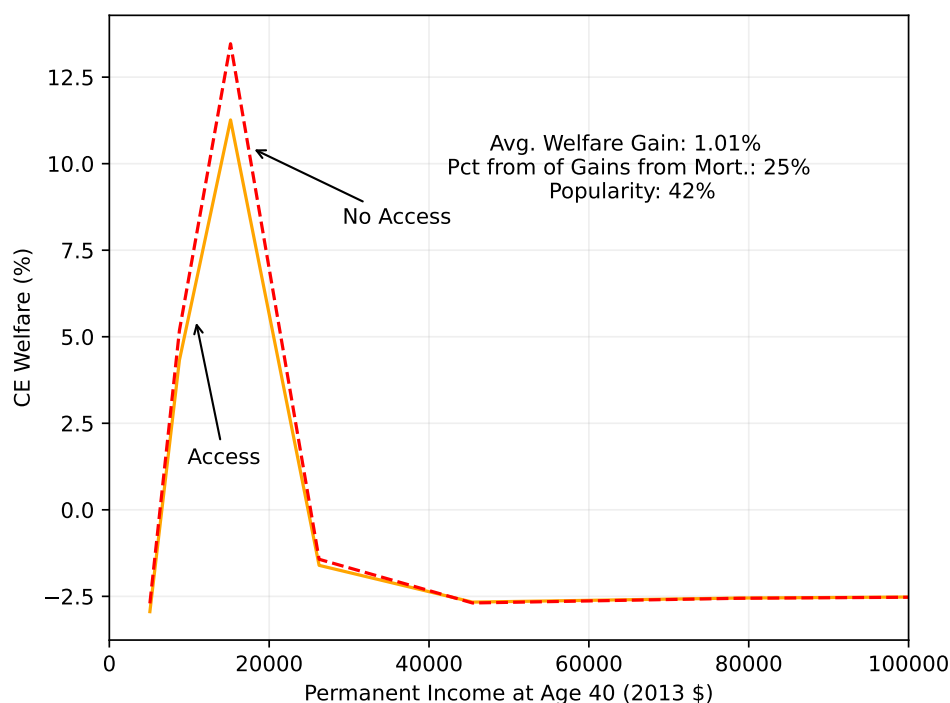
5.2. Welfare

Even if the increase in cost-effectiveness and reduction in mortality oppose each other in terms of cost savings, they both represent changes that improve welfare for recipients. Further, expansion results in substantial redistribution which, with diminishing marginal utility and a utilitarian welfare criterion, will also improve average welfare. To examine this, Figure 9 displays the consumption equivalent welfare change for newly born (i.e. age 18) individuals as a function of the permanent income and whether they are born with (solid red line) or without (dashed orange line) access to employer-provided insurance. It is important to note that this welfare comparison is being made via the general equilibrium counterfactual described above (so that no resources are being created or destroyed on net) and by comparing outcomes for newborn agents born into the pre- and post-expansion steady-states.²¹

point of overturning the main result, so that expansion actually increases costs overall. I discuss this in more detail in Appendix B.4.

²¹Outcomes for agents born anywhere along the transition path between the pre- and post-expansion steady-states would look essentially identical, largely due to the fact that changes in prices are extremely small (e.g. the interest rate increases by only 8 basis points) so that the model essentially arrives at the

Figure 9: Welfare for Newborn Individuals by Permanent Income



Note: This figure displays the change in welfare, measured in consumption-equivalence, for newborn individuals (i.e. age 18) as a result of public health insurance expansion as a function of an individual's permanent income. The solid orange line displays welfare for individuals born with access to employer-provided insurance while the dotted red line displays the same for individuals born without access.

This figure highlights the heavily redistributive nature of the policy. A small portion of the population, those who gain access to Medicaid, experiences very large gains on the order of 10 to 13 percent of consumption. The middle and top of the income distribution experience small losses of about 2.5 percent as a result of paying higher taxes yet being extremely unlikely to ever receive Medicaid. The bottom of the income distribution also loses due to higher taxes and the fact that they qualified for Medicaid even before expansion and thus gain nothing. This is reflected in the popularity of the policy; only 42 percent of the population experience a welfare gain from expansion while the remaining 58 percent experience welfare losses. Still, the gains to the winners are sufficiently large to outweigh these dispersed losses and total welfare increases by roughly 1 percent (according to a utilitarian objective function).

In order to separate these welfare gains into the impact of higher consumption and the impact of longer life expectancy, I set the preference parameter governing the utility gained from living to see each period $\bar{u}$ to zero and recompute the welfare gains (with new steady-state after only a few periods).

out allowing reoptimization). This removes the direct utility gains from longevity and leaves behind only the gains from higher consumption (more precisely, the gains from higher consumption *and* higher probabilities of living to see that consumption). The difference between this value (0.75 percent) and the total gains (1.0 percent) describes the increase in welfare that comes directly from improved longevity. Overall, the direct utility gained from reduced mortality accounts for 25 percent of the total welfare gains.

5.3. Targeting Delayed Care Directly

While the results above indicate that impact of delayed care is substantial but ultimately results in only modest cost-savings from expansion, there are two things suggesting that a policy tailored more directly towards addressing delayed care could perform better. First, Figure 5 showed that in addition to reducing delayed care, expansion created incentives for some individuals to over-consume healthcare while approaching age 65. Given that the optimal utilization profile should be smooth across the threshold, introducing this extra distortion may compromise the cost-effectiveness channel. Second, Figure 6 shows that delayed care occurs even for individuals near the median level of income (i.e. it is mostly a response to the sudden decline in the marginal cost of healthcare that occurs at age 65 rather than a result of credit-constraints impacting only the poorest individuals). Public insurance expansion (at least of the magnitude studied here) does not extend to these individuals and thus has no impact on a significant portion of the delay that occurs.

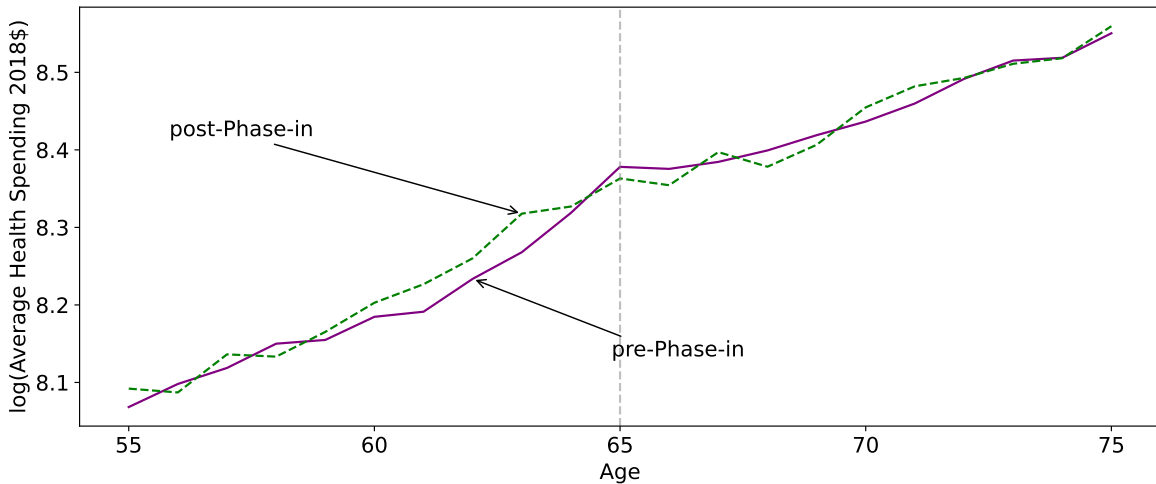
A natural policy to consider is one that slowly phases in public insurance for the elderly to avoid such any sharp changes in the marginal cost of care. Interestingly, policies along these lines do exist; for example, Japan's public insurance scheme featured a coinsurance rate that declined from 30 percent for the young to 10 percent for the elderly (see [McGrattan et al. 2018](#), for a useful discussion) — although rates for the elderly have since been raised for budgetary reasons.

To accomplish this slow phase-in, I consider a policy in which the government pays for a portion of near-old individuals' healthcare expenditure while the remaining expenditure is paid for by the individual and/or their insurance. In particular, I focus on individuals ages 61 - 64 and set rates so that each year closes 20 percent of the gap between their current cost of care and the cost of care under Medicare. As an example, an uninsured individual faces a copay rate of 100 percent at age 60 and will face the Medicare copay rate of 20 percent at age 65. Under this new policy, the government pays for 16 percent ($= \frac{1-0.2}{5}$) of their healthcare costs at age 61, 32 percent at 62, and so forth, so that their marginal cost of care varies smoothly as they age in to Medicare. For completeness, I also extend the policy to insured individuals. Given, however, that

estimated copayment rates under private insurance are similar to Medicare rates (20 percent for employer-provided and 25 percent for marketplace), the quantitative contribution of this component of the policy is negligible.²²

Results of the Phase-In Policy: Figure 10 displays the log of average healthcare spending as a function of age in the model without (purple solid line) and with (dotted green line) the phase-in policy. It is clear from the figure that the phase-in policy substantially reduces delayed care — spending from ages 60 to 64 is higher (by 3 percent on average), spending at age 65 is lower (by 2 percent), and spending after the threshold is as well (by 1 percent from 65 to 70). The result is that overall profile of spending is much smoother across the age 65 threshold.

Figure 10: Model Implied Delayed Care With and Without Phase-In



Note: This figure displays the log of average health spending. The purple line shows spending from the pre-expansion steady-state of the calibrated model while the dotted green line shows spending after the phase-in policy is implemented.

The phase-in policy spends about 0.03 percent of GDP (6 billion USD) per year; however, as was the case with the baseline expansion policy, this increase in outlays is offset to some extent by a reduction in Medicare spending. For the phase-in policy, this offset is large; for every \$100 spent on the policy, Medicare costs decrease by \$93. The policy comes very close to paying for itself, and the net cost of the policy is only 7 percent (400

²²To extend the policy to insured individuals, I consider the total (pre-insurance) government copayment/coinsurance rate needed to smoothly close the gap between the insured rates and the Medicare rate. For example, marketplace insurance exhibits a copay rate of 25 percent. To smoothly approach Medicare's rate of 20 percent, the copay rate faced by the individual must fall by 1.25 pp each year. This is accomplished by having the government pay an additional 5 percent ($= .0125/.25$) of expenditure per year so that, for example, the effective copayment rate of a 61 year old with marketplace insurance is 23.75 percent ($= 0.95 * 0.25$). To avoid distorting the incentive to purchase insurance, I also have the government pay the same percentage of insurance premiums.

million USD) of the apparent cost. As a result, the policy is more cost effective than the baseline expansion policy, saving more lives per dollar spent; mortality falls among 60-70 year olds by 0.1 percent per 230 million USD spent on the phase-in policy compared to 0.1 percent per 640 million USD spent on expansion.

The fact that the phase-in policy (which targets only the near old) is highly cost-effective while the gains from overall expansion (which targets all ages) are modest suggests that health spending by the young, far from the age 65 threshold, is somewhat inconsequential for delayed care. From this perspective, one reason for the modest gains in the main model exercise is that health spending by the young has little impact on their eventual health at age 65, and expansion thus directs a lot of resources towards unimportant (from the perspective of delayed care) spending. The phase-in policy, on the other hand, is better targeted. One of the sensitivity analysis in Appendix Table [A.4](#) tests this intuition directly. Increasing the rate of health depreciation (fifth column) increases the dependence of late-in-life health on early-in-life spending; average health for those younger than 55 with and without employer-provided health insurance is 0.96 and 0.90 respectively under baseline parameters, but the difference between the two nearly doubles (0.94 vs 0.84) when depreciation is increased. As expected, this results in higher cost effectiveness for the baseline policy (\$17 saved per \$100 spent) and lower effectiveness for phase-in (\$39 per \$100) — under this alternative parameterization, the phase-in policy's advantage in targeting is less important, shrinking the gap between the two.

Overall, the results of this exercise suggest that while the baseline expansion policy does reduce the extent of delayed care, it does so fairly inefficiently due to poor targeting on a number of dimensions. The phase-in policy targets delayed care among the near-elderly directly and leads to substantially higher cost-savings. It is important to note, however, that this policy is very limited in scope. Although it could, in theory, be expanded by further lowering the age threshold at which coverage begins, the threshold of 60 is already sufficient to achieve almost all the potential gains from reducing delayed care, and lowering it further does not meaningfully impact outcomes. Thus the potential scale for such a policy is inherently limited.

6. Conclusion

Evidence suggests that a substantial number of U.S. individuals delay healthcare until they receive health insurance through Medicare at age 65. This paper provides a quantitative model to explain this fact and analyze the extent to which public health insurance expansion can reduce individual incentives to delay care. A key question is the extent to which reductions in delayed care lead to cost savings, due to earlier

care being more effective than late care, and how much of these savings are offset by cost increases due to reductions in mortality. To discipline these channels, I combine an existing quasi-experimental result from the health literature with my own empirical analysis. In particular, the cost-effective channel is disciplined using the jump in healthcare consumption that occurs in the month that an individual turns 65, as measured using the MEPS, and the decline in mortality is disciplined using the 2014 ACA Medicaid expansion diff-in-diff results from [Miller et al. \(2021\)](#).

My results suggest that the overall costs savings are modest; Medicare costs decline by only \$8 for every \$100 spent on expansion. This small overall decline arises from the fact that although the impact of the cost-effectiveness channel is large (\$31 per \$100 spent), the increase in costs due to declining mortality serves to offset almost three-quarters of these savings.

Despite the increase in insurance coverage, there remains a substantial amount of delayed care even after insurance access is expanded — a result of the fact that delaying care is a substantial phenomenon even at the median level of income and is not limited to just the poor and credit-constrained. Consequently, the baseline expansion policy is poorly suited to address the issue of delayed care. A phase-in policy that gradually introduces insurance coverage for the elderly is better suited for this purpose, almost completely eliminates delayed care, and leads to substantial reductions in Medicare costs equal to \$93 per \$100 spent on the policy. As a result, this policy is substantially more cost-effective than the baseline expansion policy, in terms of mortality reduction per dollar spent.

One key limitation of this analysis is that it assumes that all medical spending is productive on the margin and that all of the observed increase in healthcare consumption at age 65 represents increased investment into individual health. If, instead, increases in consumption at age 65 are a mix of necessary, productive care and of physicians prescribing unnecessary, unproductive care to exploit Medicare reimbursement rules, my analysis will overestimate the importance of delayed care. For example, empirical work has argued that some prescribed treatments are overused and have very little or even negative value ([Kowalski 2021](#), is one example for the case of mammograms). It is unclear how this would translate into my central results; such a phenomenon would mean that the model simultaneously over-predicts the cost-savings due to the reduction in delayed care but under-predicts the reduction in mortality and the resulting cost increase, pushing in opposite directions on the baseline number of \$8 saved per \$100 spent. Further analysis that explicitly models the decision problem of physicians and leverages solid evidence on the extent of over-treatment would be valuable.

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A. Additional Figures and Tables

Table A.1: MEPS Medical Provider Survey Coverage

Provider	Coverage
Hospitals	100%
Office-based Physicians	100% (Medicaid + Medicare covered individuals) 75% (HMO or managed care covered individuals) 25% (remaining individuals)

Note: This table reports the percentage of medical spending covered by the Medical Provider component of the Medical Expenditure Panel Survey for 4 different categories of care. Information taken from [Sommers \(2007\)](#).

Table A.2: Some Summary Statistics of Investment and Emergency Spending

	All			65 or Older		
	Mean	Median	% > 0	Mean	Median	% > 0
Investment	\$2,548	\$440	89%	\$4,945	\$2,503	97%
Emergency	\$1,543	\$0	18%	\$3,382	\$0	25%

Note: This table provides some summary statistics for healthcare investment and emergency spending. Emergency spending is defined as any spending that takes place in the emergency department and investment spending is all other spending. Calculated from the Medical Expenditure Panel Survey year 2013

Table A.3: Estimated Insurance Plan Parameters

Plan	Copay Rate	Deductible	Coins. Rate	Premium
Employer	0.20	\$1,550	0.06	Deter. in Eq.
Marketplace	0.25	\$1,550	0.13	Deter. in Eq.
Medicaid	0.03	\$0	0.03	\$0
Medicare	0.20	\$1,200	0.0	\$1,250
Uninsured	1.0	\$0	1.0	\$0

Note: This table displays the calibrated and estimated insurance plan parameter. See discussion for details

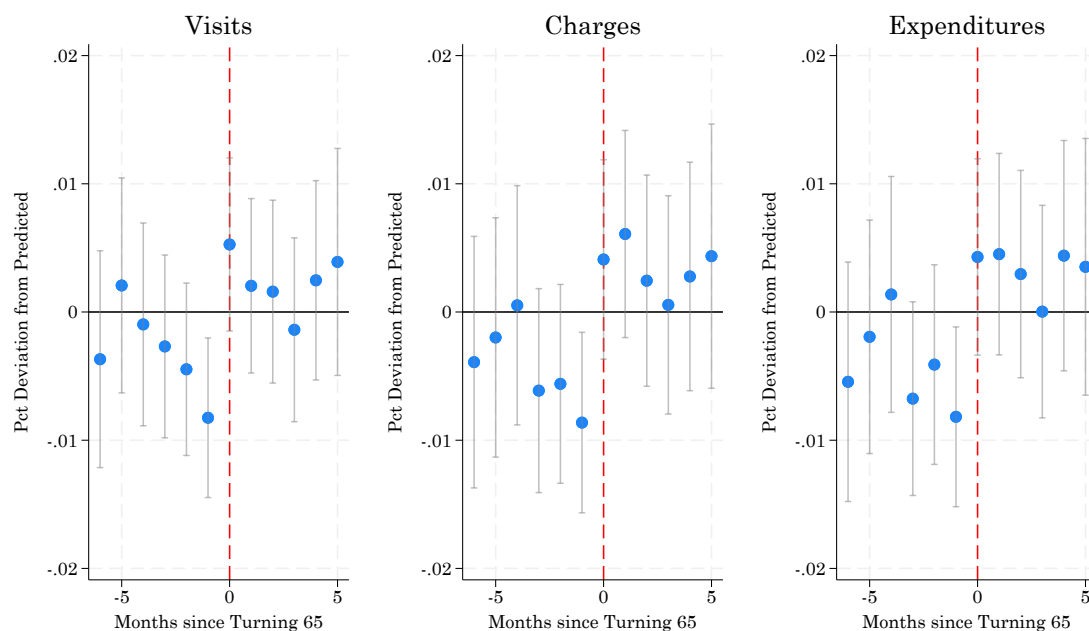
Table A.4: Sensitivity Analysis of Main Results

	Baseline	$\uparrow \pi_h$	$\uparrow \phi$	$\uparrow \psi$	$\uparrow \delta_a$	$\uparrow \phi, \uparrow \psi, \uparrow \delta_a$
MCD Spend. (% of GDP)	+0.31	+0.32	+0.24	+0.38	+0.42	+0.52
MCR Spend. (% of GDP)	-0.02	-0.01	-0.01	-0.01	-0.07	+0.02
MCR Savings per \$100	\$8	\$3	\$3	\$1	\$17	-\$3
Savings per \$100 (Phase-in)	\$93	\$33	\$24	\$35	\$39	-\$26

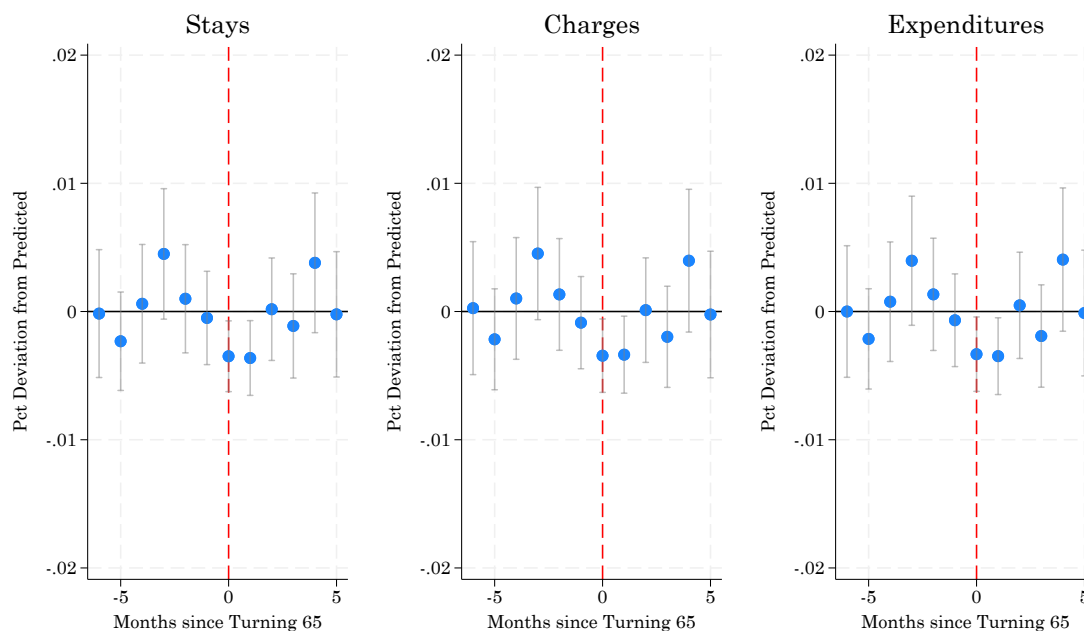
Note: This table displays the sensitivity of the main results to variation in model parameters. All single parameters are increases by 25 percent, excepting ϕ which is increase by 10 percent (to avoid drifting too far from the targeted level of average health spending). The final column increases ϕ and ψ by 100 percent and δ_a by 50 percent (also to keep average health spending near the targeted value).

Figure A.1: Health Utilization by Months since Turning 65

(a) Office-Based Visits

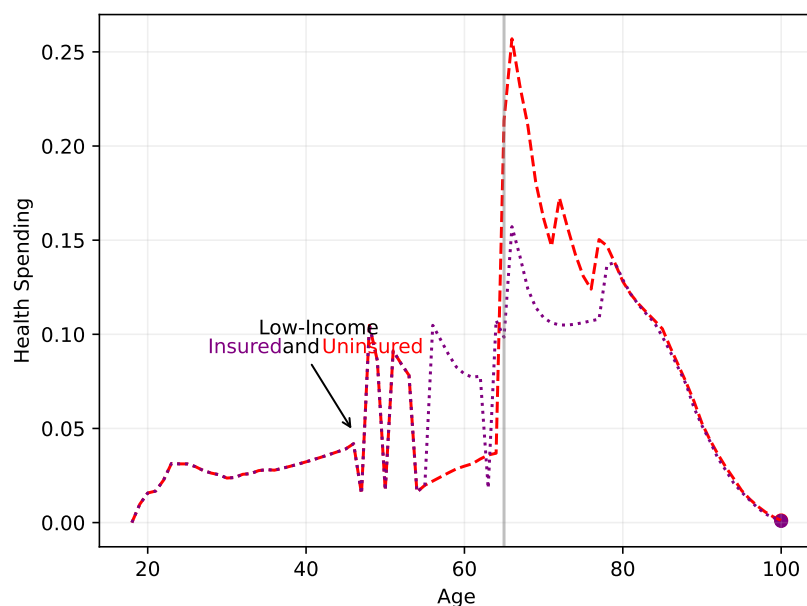


(b) Hospital Inpatient Stays



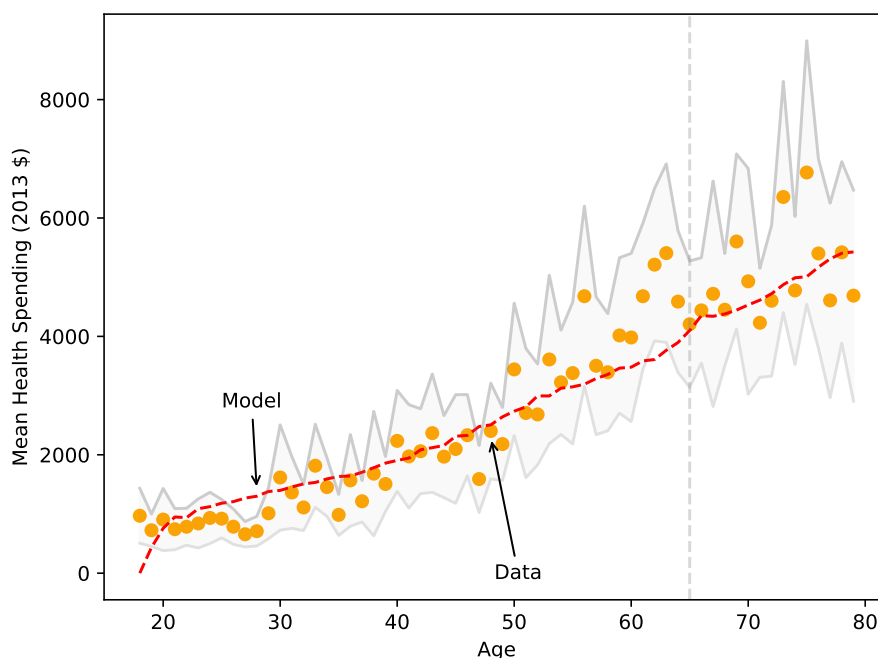
Note: This figure displays the percent deviation from the predicted value under a uniform distribution (described further in the text) for six measures of healthcare utilization, three for office-based-provider visits and three for hospital inpatient stays, as a function of the months since an individual turned age 65.

Figure A.2: Life-cycle Health Spending for Low-Income Insured and Uninsured



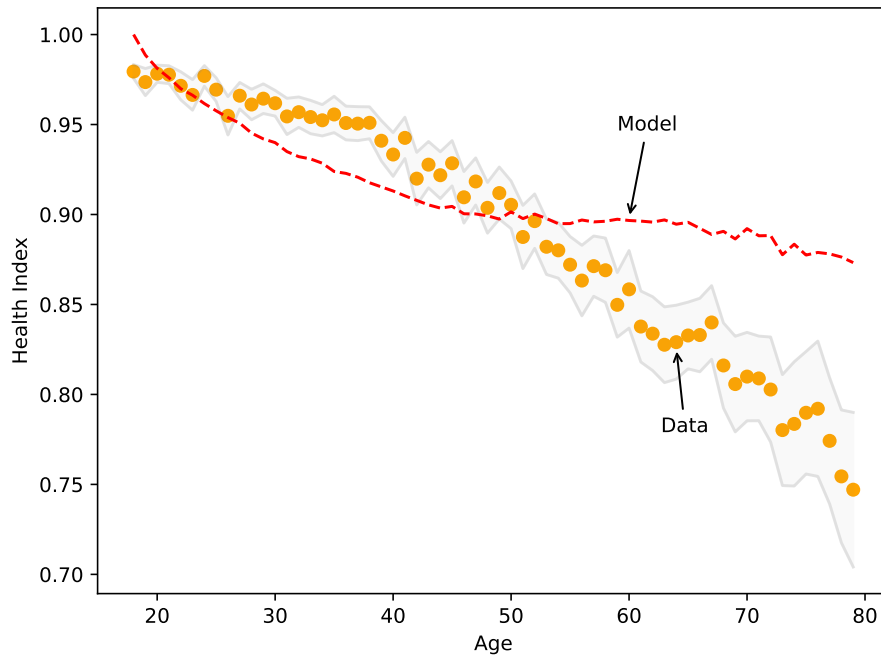
Note: This figure displays life-cycle health spending for two individuals. The purple line displays the spending for an individual with low permanent income and access to employer based insurance. The red line displays spending for an individual with low permanent income and no access to employer-based insurance.

Figure A.3: Spending vs Age: Data and Model



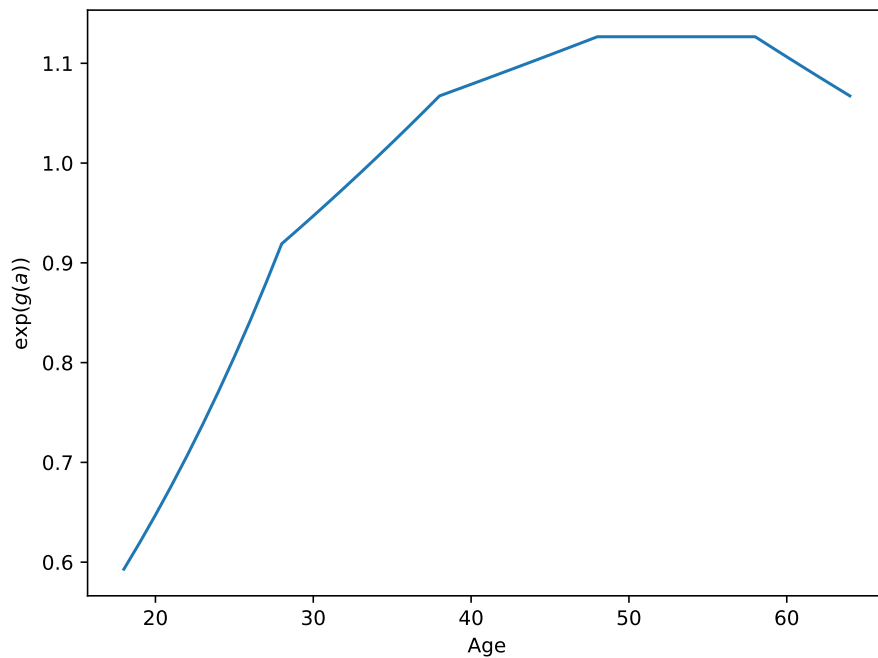
Note: This figure displays average (log) individual health spending as a function of age in both the estimated model and as measured in the 2013 wave of the MEPS.

Figure A.4: Health vs Age: Data and Model



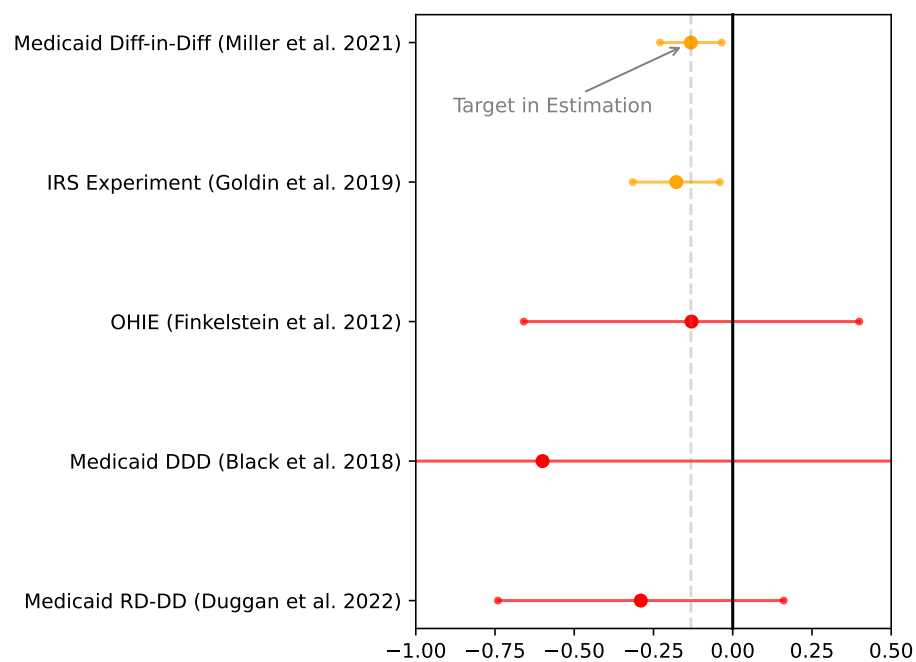
Note: This figure displays individual health h as a function of age in both the estimated model and as measured in the 2013 wave of the MEPS.

Figure A.5: Calibrated Lifecycle Component of Income



Note: This figure displays the calibrated life-cycle component of income taken from [Lagakos et al. \(2018\)](#).

Figure A.6: The Impact of Health Insurance Coverage on All-Cause Mortality



Note: This figure plots the point estimates and 95% confidence intervals for the impact of health insurance coverage on all-cause mortality for five different studies from the health economics literature. Estimates that are statistically significant at the 5% level are in orange while those that are not are in red.

B. Miscellaneous Model Notes

This section details various model notes and extensions.

B.1. Time Inputs into Health

A natural model extension is to allow a time input into health formation in addition to the monetary input. Exercise is the most natural interpretation of such an input, but it could also be thought of as summarizing the time that must be spent gathering information, handling administrative tasks, and traveling or waiting to receive medical care. Not only are these costs potentially important, including them is necessary to apply the model to countries where the marginal monetary cost of medical care is close-to-zero (to the consumer).

There are many possible ways to add this extension, but two natural assumptions allow the time input to be included without additional computational cost. Even stronger, these assumptions imply that the model with time inputs is isomorphic to the model without time inputs, justifying the decision to leave them out of the baseline model.

The first assumption is that the time input to health, henceforth e_t for “exercise”, experiences the same decreasing returns to scale as the goods input i_t . The assumption is essential as it ensures that exercise is neither a luxury nor necessity good — without it, the “share” of exercise as an input to health would be rising or falling with income. There are many ways to ensure this, but the simplest is to add an additional term so that the health accumulation equation is now given by

$$h_{t+1} = (1 - \delta_a - \delta^x)h_t + \phi_i i_t^\psi + \phi_e e_t^\psi \quad (17)$$

where ϕ_i corresponds to the model parameter ϕ but may take a different value in the model with exercise (making this distinction facilitates discussion of the isomorphism in a few paragraphs).

The second assumption is that agents’ preferences take the time input into health as a perfect substitute for the aggregate labor input — period utility takes the form $u(c_t, l_t + \alpha_e e_t)$. This assumption ensures that the marginal benefit of additional exercise, like the marginal benefit of additional medical spending, has a linear relationship with the marginal utility of consumption (through the labor-leisure condition).

Combining these two assumptions and solving for agents optimal choice of e_t yields

$$e_t^* = \left(\frac{\phi_e}{\alpha_e w_t \phi_i} \right)^{\frac{1}{1-\phi}} i_t^* \quad (18)$$

where w_t is the “wage per unit of labor aggregate” implied by the cost-minimization problem in (28). Plugging this into the health accumulation equation (17) and rearranging yields

$$h_{t+1} = (1 - \delta_a - \delta^x)h_t + \phi_i \left(1 + \left(\frac{1}{\alpha_e w} \right)^{\frac{\psi}{1-\psi}} \left(\frac{\phi_e}{\phi_i} \right)^{\frac{1}{1-\psi}} \right) i_t^\psi \quad (19)$$

The isomorphism arises from the fact that for any parameter ϕ from a model without a time input to health and target time-input-to-goods-input ratio γ , choosing ϕ_i and ϕ_e subject to

$$\phi = \phi_i \left(1 + \left(\frac{1}{\alpha_e w} \right)^{\frac{\psi}{1-\psi}} \left(\frac{\phi_e}{\phi_i} \right)^{\frac{1}{1-\psi}} \right) \quad (20)$$

$$\gamma = \left(\frac{\phi_e}{\alpha_e w_t \phi_i} \right)^{\frac{1}{1-\phi}} \quad (21)$$

generates a model that matches aggregate exercise share γ and generates identical health behaviors as the model without the time input.

B.2. The Value Function and Recursive Competitive Equilibrium

The individual faces eight individual-level state variables. They are

1. Assets b
2. Health h
3. Age a
4. Permanent productivity z^p
5. Temporary productivity z^s
6. Insurance plan p
7. Access to employer-provided insurance e
8. Information status χ

They also face an aggregate state variable Ω describing the cross-sectional distribution of all individuals across the 8 individual-level states. The individual problem for a well-informed individual not bound by the consumption floor can be written recursively as in (12). $G(\Omega)$ is the perception function used by the individual to forecast the future aggregate state. The problem for a poorly-informed individual is similar but replaces

the actual health-related stochastic processes $\pi(h, a)$, $\pi_x(h, a)$, $\mu(h, a)$, and $\sigma(h, a)$ with their perceived counterparts $\pi(h^*, a)$, $\pi_x(h^*, a)$, $\mu(h^*, a)$, and $\sigma(h^*, a)$ for $h^* = (1 - \chi)h + \chi$.

$$\begin{aligned}
V(b, h, a, z^p, z^s, p, e; \Omega) = & \max \bar{u} + u(c, l) + \\
& + \beta(1 - \pi(h, a))\mathbb{E}[V(b', h', a + 1, z^p, z^{s'}, p', e'; \Omega')] \\
s.t. \quad c + b' + \chi_p(i, m) = & (1 + r(\Omega))b + T(w(\Omega)z(z^p, z^s, a)l) \quad \text{if } a < 65 \\
c + b' + \chi_{\text{MCR}}(i, m) = & (1 + r(\Omega))b + y_{a \geq 65}(z^p) \quad \text{if } a \geq 65 \\
h' = & (1 - \delta_a - \delta_x)h + \phi i^\psi \\
p' \in & \{\text{EMP, IND, UN, MCD}\} \text{ according to eligibility} \quad (22)
\end{aligned}$$

$$b' \geq 0$$

$$i \geq 0$$

$$z^{s'} \sim \rho z^s + \varepsilon, \quad \varepsilon \sim N(0, \sigma)$$

$$m \sim \begin{cases} 0 & \text{with prob. } 1 - \pi_x(h, a) \\ \log N(\mu(h, a), \sigma(h, a)) & \text{with prob. } \pi_x(h, a) \end{cases}$$

$$e' \sim M(e)$$

$$\Omega' = G(\Omega)$$

An individual bound by the consumption floor solves

$$\begin{aligned}
\underline{V}(b, h, a, z^p, z^s, p, e; \Omega) = & \max \bar{u} + u(c, l) + \\
& + \beta(1 - \pi(h, a))\mathbb{E}[V(0, h', a + 1, z^p, z^{s'}, p', e'; \Omega')] \\
s.t. \quad c = & \underline{c} \quad (23)
\end{aligned}$$

$$\begin{aligned}
c + b' + \chi_p(i, m) = & (1 + r(\Omega))b + T(w(\Omega)z(z^p, z^s, a)l) + X_{\underline{c}} \quad \text{if } a < 65 \\
c + b' + \chi_{\text{MCR}}(i, m) = & (1 + r(\Omega))b + y_{a \geq 65}(z^p) + X_{\underline{c}}, \quad \text{if } a \geq 65 \\
h' = & (1 - \delta_a - \delta_x)h + \phi i^\psi \quad (24)
\end{aligned}$$

where $X_{\underline{c}}$ is equal to $\underline{c} + b' + \chi_p(i, m) - ((1 + r(\Omega))b + T(w(\Omega)z(z^p, z^s, a)l))$ for those under age 65 and $\underline{c} + b' + \chi_{\text{MCR}}(i, m) - ((1 + r(\Omega))b + y_{a \geq 65}(z^p))$ for those 65 and over, but is perceived as constant by the individual.

The recursive competitive equilibrium of the model consists of

- a) Individual value and policy functions for both good-information and bad-information individuals given by (V, c, i, b', l, p') and $(V_{\chi}, c_{\chi}, i_{\chi}, b'_{\chi}, l_{\chi}, p'_{\chi})$
- b) Firm policy functions (L, K, Y)
- c) Price functions $(r, w, P_{\text{EMP}}, P_{\text{IND}})$
- d) A perception function G
- e) A tax-level parameter λ_{τ}

such that

- 1) The value and policy functions in a) solve the individual optimization problem (12)
- 2) The firm policy functions solve the firm optimization problem:

$$\max AL^{\alpha}K^{1-\alpha} - rK - wL$$

- 3) Markets clear: $\int bd\Omega = K$, $\int ld\Omega = L$, and (7) and (8) hold
- 4) Perceptions are correct: $\Omega' = G(\Omega)$
- 5) Government budget given in (15) is balanced

B.3. Expansion in General Equilibrium

One could worry that the simplifying assumption of a single sector producing both consumption and healthcare goods, essentially fixing their relative prices in a partial equilibrium sense, could have a large impact on the baseline results. In particular, if expansion increases the relative price of healthcare goods and, consequently, increases the price of insurance, it could be the case that some of the reduction in delayed care that occurs due to higher rates of Medicaid coverage is undone by higher uninsurance rates elsewhere in the income distribution. Here I extend the baseline model to include such general equilibrium effects. This extension is used to test the robustness of the baseline results in Table 3.

Production for consumption and healthcare goods now occurs via two distinction production functions given by

$$\begin{aligned} p_h Y_h &= p_h A_h L_h^\alpha K_h^{1-\alpha} \\ Y_c &= A_c L_c^\alpha K_c^{1-\alpha} \end{aligned} \quad (25)$$

where p_h is the price of healthcare good (consumption is the numeraire). The budget constraint for individuals is adjusted to include the price of healthcare as well. Capital can flow freely between the two sectors so that $K = K_c + K_h$, and there remains a single effective interest rate r .

Individuals now much decide how much labor to supply to each sector. I opt to model this in a simple, reduced-form way. An individual who supplies labor l_h to the healthcare sector and labor l_c to the consumption goods sector earns pretax income given by

$$y_{\text{pre-tax}} = (w_h l_h + w_c l_c) z(z^p, z^s, a) \quad (26)$$

Individuals face disutility from their aggregate labor supply l which is given by a CES-style aggregator of labor supply to the healthcare and consumption sectors

$$l = \left((1 - \alpha_h) l_h^{\frac{\xi+1}{\xi}} + \alpha_h l_c^{\frac{\xi+1}{\xi}} \right)^{\frac{\xi}{\xi+1}} \quad (27)$$

While abstract, this reduced-form description of labor supply captures an upwards-sloping relative supply curve for healthcare labor in a tractable way by allowing the relative labor supply decision to be solved analytically, yielding a relative labor supply towards the health sector $\frac{l_h}{l_c}$ that exhibits a constant elasticity of ξ with respect to the relative wage. Conditional on aggregate labor supply l , the labor allocation problem of the individual is

$$\begin{aligned} \max \quad & w_h l_h + w_c l_c \\ \text{s.t.} \quad & l = \left((1 - \alpha_h) l_h^{\frac{\xi+1}{\xi}} + \alpha_h l_c^{\frac{\xi+1}{\xi}} \right)^{\frac{\xi}{\xi+1}} \end{aligned} \quad (28)$$

which yields relative labor supply to the medical section given by

$$\frac{l_h}{l_c} = \left(\frac{1 - \alpha_h}{\alpha_h} \frac{w_h}{w_c} \right)^\xi \quad (29)$$

Thus the relative supply of labor to the medical sector $\frac{l_h}{l_c}$ exhibits a constant elasticity of ξ with respect to the relative wage $\frac{w_h}{w_c}$. I also allow α_m to vary with respect to permanent

income z_p in order to match the relationship between income and the propensity to work in healthcare as measure in the American Community Survey.

The value for the elasticity of substitution ξ is backed-out from the quasi-experimental results of [Finkelstein \(2007\)](#) who uses a difference-in-difference framework exploiting pre-existing differences in elderly insurance coverage and the national implementation of Medicare to estimate the effect of Medicare on aggregate healthcare outcomes, including hospital employment and hospital payroll. She finds that Medicare increased employment by 25.6 percent and payroll by 40.1 percent in the 5-10 years following implementation (the longest-run time horizon available). The results suggest that earnings per worker increased by 11.5 percent. I treat both the increase in employment and the increase in earnings per worker as increases in the relative employment share and relative wage for healthcare yielding an elasticity of approximately 2.22 ($= \frac{.256}{.115}$).

The final aspect of general equilibrium is to account for the increase in taxation necessary to fund expansion. In estimation of the baseline model, λ_τ is chosen to match a steady-state tax to GDP ratio of 24 percent, and the value of G is then taken as the difference between total revenues and these other forms of expenditure so that the government budget constraint (15) holds exactly at the chosen value of λ_τ . In this extension to general equilibrium, the value of G is fixed at the value constructed during model estimation, and the value of λ_τ is adjusted in order to maintain government budget balance. Thus the new post-policy equilibrium is the fixed point in prices and λ_τ where markets clear and the government budget constraint holds exactly.

The general equilibrium exercise reported in [Table 3](#) begins from the same stationary steady-state as the baseline model (i.e. the relative price of healthcare is normalized to unity). After expansion, I compute the change in p_h , change in insurance premiums, change in interest rate, and change in taxation implied by equilibrium in this new model. Comparisons are then made between this post-expansion steady-state and the price-expansion steady-state.

B.4. Discussion of [Appendix Table A.4](#)

[Appendix Table A.4](#) reports the results of sensitivity analysis aimed at answering two related questions. The first is whether the model's conclusion that public health insurance expansion leads to modest cost-reductions by reducing delayed care is simply "baked-in" to the model structure or whether it is the result of the exact parameter values estimated via SMM and, consequently, the targeted moments. The second is simply how robust the model's main conclusions are to reasonable change in the values of key parameters.

The results in this table are constructed by starting from the baseline estimated pa-

parameter values and increasing a key parameter value. The values I focus on are the derivative of mortality with respect to health (π_h), the productivity of health spending (ϕ), the returns to scale of health spending (ψ), and the depreciation rate of health (δ_a , increase proportional at all ages). All parameters are increased by 25 percent, excepting ϕ which is increased by 10 percent (otherwise average health spending becomes too low to make a reasonable comparison). In the final column, I increase ϕ , ψ , and δ_a simultaneously by 100, 100, and 50 percent respectively with these values being chosen so that the value of aggregate moments are more-or-less than same as the estimated steady-state.

Increasing the impact of health on mortality and the productivity of health spending both increase the strength of the mortality channel, as spending and coverage become more important, and thus reduce the scope for potential gains and the costs savings from expansion, as well as the phase-in policy targeting delayed care directly from Section 5.3. Similarly, increasing the returns to scale of health spending ψ shrinks the size of the cost-effectiveness channel, as delayed care can more easily be made up for by later spending, and also decreases the cost savings in both cases. The effect of increasing depreciation is discussed in more detail in Section 5.3.

The most interesting result is the final column, which simultaneously increases the productivity of health spending, the returns to scale, and depreciation. All three of these changes increase the strength of the mortality channel and, together, increase it by enough to overturn the model's core result that expansion leads to long-run cost savings. In this alternative model, both the baseline expansion policy and the phase-in policy actually *increase* eventual Medicare costs due to such a large improvement in survival rates. This result underscores the fact that the model's conclusions are not forgone and instead depend crucially on the estimate parameters and, consequently, the targeted moments.

C. Details on Estimation and Estimated Model Parameters

This subsection fills in the model estimation details missing from the discussion in 4.1. Table C.1 lists the model parameters taken from external sources or set to standard values. The persistence and variance parameters of the income process are taken from [Floden & Lindé \(2001\)](#). The life-cycle component of income is taken from [Lagakos et al. \(2018\)](#). The functional form for the tax function as well as the progressivity parameter τ are taken from [Heathcote et al. \(2017\)](#) who estimate this simple functional form as a continuous and smooth approximation to the US income tax schedule.

Table C.1: Externally Calibrated Parameters

Description	Parameter	Value
Discount Factor	β	0.97
Utility from (c, l)	$u(c, l)$	$\frac{1}{1-\sigma} c^{1-\sigma} (1 - \kappa(1 - \sigma) l^{1+\frac{1}{\nu}})^{\sigma}$
Coefficient of Relative Risk Aversion	σ	2
Frisch Elasticity of Labor	ν	1
Disutility of Labor	κ	0.3
Income Persistence	ρ	.91
Income SD	σ	.04
Life-cycle Income	$g(a)$	See Figure A.5
Labor Share	α	0.66
Healthcare Labor Supply Elasticity	ξ	2.22
Tax Function	$T(y)$	$\lambda_{\tau} y^{1-\tau}$
Tax Progressivity	τ	0.181
Social Security Function	$y_{a \geq 65}(z_p)$	See discussion

Note: This table displays model parameters calibrated to standard literature values. See discussion for details on each parameter.

Table C.2 lists the parameters of the model that are estimated directly or close-to-directly from data, their values, and the data source on which they are estimated. The distribution of individual-level permanent income is chosen to be log-normal with the mean and variance parameters calibrated to match US GDP per capita and median per-

sonal income.²³ The effective loading factors $(1 - s_{\text{EMP}})\kappa_{\text{EMP}}$ and κ_{IND} are calculated as the ratio of total premiums paid over total covered costs for all individuals in the MEPS covered by employer-provided and marketplace insurance respectively.

The parameters for the representative insurance plans (summarized in Appendix Table A.3) are constructed by computing the portion of total of spending in a category that comes out-of-pocket. For example, the copayment rate for employer-provided insurance is estimated by dividing the total amount of out-of-pocket non-emergency spending by the total amount of non-emergency spending among those with employer-provided insurance. Coinsurance rates and rates for marketplace and Medicaid insurance plans are estimated similarly (with the caveat that coinsurance rates must account for deductibles). The parameters for the Medicare insurance plan are set to their statutory values. Loading factors κ_{EMP} and κ_{IND} are estimated similarly by dividing total premiums collected by total outlays for by employer-provided and marketplace-purchased plans.

The Markov process for the availability of employer-provided insurance, and how it varies with permanent income z_p , is chosen to match two regressions from MEPS data. The first is a regression of the probability of being covered via EPHI as a function of (log) family income, which yields a slope coefficient of 0.23 (and intercept 0.60 to match the overall mean). The second is a regression of the probability of losing access to EPHI, conditional on having access, as a function of (log) income which yields a slope coefficient of 0.002 (though this is statistically insignificant). The exogenous age-premium profile $p(a)$ is estimated directly from MEPS data on privately insured individuals ages 25-65 via linear regression of premiums on age and the square of age.

The mortality and emergency expenditure functions are all estimated directly from individual-level data on age, health, mortality, and emergency expenditure in the MEPS. All four are estimated using OLS with individual health, age, and the square of age as regressors.²⁴ In the case of the variance of emergency expenditure, I construct the individual-level variance for each observation as the squared residual from the regression used to estimate mean emergency expenditure. I then regress this individual-level variance on the predictors which recovers the best linear predictor of $\mathbb{E}([Y_i - \mathbb{E}(Y_i|X_i)]^2|X_i)$ where Y_i is emergency expenditure and X_i are the predictors which is ex-

²³Using the mean and median as targets rather than a moment summarizing the tail behavior of the income distribution such as the top 1 percent income shares reflects the fact that expansion of public health insurance is a policy that mostly affects low- and middle-income individuals.

²⁴Using OLS even for binary outcomes like mortality ensures that the estimated marginal benefit of health is constant (or monotonic if the square of health were included) everywhere. This is not necessarily true in logit or probit estimation, and any non-monotonicities would substantially complicate computation of the model.

Table C.2: Directly Estimated Parameters

Description	Parameter	Value	Data
Mortality Function	$\pi(h, a)$	Table C.3 Col. 1	MEPS
Emerg. Prob. Function	$\pi_x(h, a)$	Table C.3 Col. 4	
Emerg. Mean Function	$\mu(h, a)$	Table C.3 Col. 7	
Emerg. Var Function	$\sigma(h, a)$	Table C.3 Col. 10	
Premium-Age Profile	$p(a)$	$-0.024 + 0.003a - 0.00003a^2$	MEPS
Pre-expansion MCD Cutoff	$\bar{z}_{\text{PRE}}$	0.15	Miller et al. (2021)
Post-expansion MCD Cutoff	$\bar{z}_{\text{POST}}$	0.25	Miller et al. (2021)
EMP availability	$M(z_p)$	$\begin{bmatrix} 1 - (.042 + .002z_p) & .042 + .002z_p \\ 1 - (.042 + .002z_p)^{\frac{.60 + .23z_p}{1 - (.60 + .23z_p)}} & (.042 + .002z_p)^{\frac{.60 + .23z_p}{1 - (.60 + .23z_p)}} \end{bmatrix}$	MEPS
EMP Load Factor	$(1 - s_{\text{EMP}})\kappa_{\text{EMP}}$	0.98	MEPS
IND Load Factor	κ_{IND}	1.43	MEPS
Insurance Plans		See Table A.3	
Perm. Income	z_p	$\log N(\mu_z, \sigma_z)$	See discussion
Tax Level	λ_τ	0.73	Tax-to-GDP of 25%

Note: This table displays model parameters estimates directly or close-to-directly from data as well as the data source or aggregate target. See discussion for details on each parameter.

Table C.3: Results of Mortality and Emergency Spending Regressions

VARIABLES	(1) Mortality	(2) Mortality	(3) Mortality	(4) Emerg. > 0	(5) Emerg. > 0	(6) Emerg. > 0
Health	-0.0544*** (0.00344)	-0.0483*** (0.00344)	-0.0217*** (0.00361)	-0.903*** (0.0404)	-0.824*** (0.0410)	-0.724*** (0.0453)
Age	-0.00177*** (9.32e-05)	-0.00177*** (9.55e-05)	-0.00190*** (9.54e-05)	-0.00744*** (0.00107)	-0.00595*** (0.00109)	-0.00853*** (0.00107)
Age ²	2.18e-05*** (1.11e-06)	2.18e-05*** (1.12e-06)	2.37e-05*** (1.14e-06)	6.06e-05*** (1.15e-05)	4.79e-05*** (1.17e-05)	7.38e-05*** (1.15e-05)
Observations	164,506	160,592	163,792	11,812	11,517	11,812
R-squared	0.028	0.028	0.036	0.071	0.089	0.078
Controls		YES			YES	
Self-Reported Health			YES			YES

VARIABLES	(7) log(Emerg.)	(8) log(Emerg.)	(9) log(Emerg.)	(10) Variance	(11) Variance	(12) Variance
Health	-1.301*** (0.348)	-1.580*** (0.360)	-0.938* (0.392)	-2.765** (0.841)	-2.604** (0.862)	-2.484** (0.962)
Age	-0.0241* (0.0117)	-0.0313* (0.0122)	-0.0273* (0.0118)	0.0431 (0.0276)	0.0424 (0.0287)	0.0411 (0.0280)
Age ²	0.000378** (0.000116)	0.000437*** (0.000120)	0.000418*** (0.000116)	-0.000412 (0.000270)	-0.000383 (0.000279)	-0.000388 (0.000274)
Observations	2,010	1,945	2,010	2,010	1,945	2,010
R-squared	0.054	0.070	0.061	0.013	0.018	0.013
Controls		YES			YES	
Self-Reported Health			YES			YES

Robust standard errors in parentheses

*** p<0.001, ** p<0.01, * p<0.05

Note: This table displays the results of regressions of various health outcomes on health and age with robustness to the inclusion of some basic controls (race, gender, family income, and census region) and a Likert measure of self-reported health. Columns (1-3) display the results of a regression of mortality on age and health with no controls, a battery of controls, and a control for self-reported measures of health respectively. Columns (4-6) display the same regression with the probability of positive emergency expenditure as the outcome. Columns (7-9) and (10-12) display the results of the same regression with log emergency spending and the variance of log emergency spending as the outcomes, for those with non-zero emergency spending. Columns (1-3) are calculated using MEPS data from 2007 to 2013, with the extra years of data being added to ensure that there are a sufficient number of observations for the rare outcome of mortality, while the remaining columns are calculated using MEPS data from 2013 only.

Table C.4: Moments Targeted by SMM

Moment	Model	Data	Source	Parameter
Avg. VSL	\$10 million	\$10 million	Viscusi & Aldy (2003)	$\bar{u}$
Jump in Medical Exp. at 65	0.9%	0.9%	Table 1	ψ
Mortality Drop from ACA	See Figure 3		Miller et al. (2021)	ϕ
Avg. Health Spending	\$2,612	\$2,450	MEPS	χ
90-10 Diff. Health Spending	\$5,955	\$6,500	MEPS	λ_χ
Avg. Health	0.92	0.91	MEPS	δ_{18}
cov(Health Spend, Age)	0.45	0.46	MEPS	δ_{100}
Emergency Health Diff.	-0.10	-0.07	MEPS	δ_e

Note: The table displays the moments targeted in the simulated method of moments estimation along with their value in both the estimated model and the data. Also displays a rough correspondence between targeted moments and model parameters. See discussion for more details.

actly the conditional variance. The procedure is very similar to performing a Breusch–Pagan test of heteroskedasticity. Columns (1), (4), (7), and (10) display the regression results for the outcomes of mortality, greater than 0 emergency expenditure, mean emergency expenditure, and the variance of emergency expenditure respectively. These are the estimated functions used in the model. The remaining columns correspond to robustness checks for additional controls.

The remaining parameters are estimated using the simulated method of moments. Table C.4 lists the targeted moments, their values in the model and data as well as the source of the data value, and their rough correspondence to model parameters. Table C.5 lists the values of the parameters themselves. The most straightforward correspondence is that between the parameter $\bar{u}$, which governs the utility an individual received for being alive each period, and individuals' value of statistical life within the model. I use a value of \$10 million from Viscusi & Aldy (2003).

As discussed in the main text, the productivity and returns to scale of health spending are pinned by the empirical results of Miller et al. (2021) and Table 1 respectively. Conditional on these moments (and the VSL), the average level of healthcare spending and the average level of health pin down the parameter governing the degree to which individuals misperceive health χ and the initial depreciation of health δ_{18} . The intuition is that, conditional on how individuals value mortality and how effective care is at pre-

Table C.5: Parameters Estimated by SMM

Parameter	Value	Parameter	Value
$\bar{u}$	310	λ_χ	0.88
ψ	0.22	δ_{18}	0.04
ϕ	0.11	δ_{100}	0.08
χ	0.5	δ_e	0.02

Note: The table displays the parameter values estimated by the simulated method of moments.

venting death, any change in health spending must be due perceptions of returns. Then depreciation governs what level of health that amount of spending achieves.

The parameter governing the share of individuals who have bad information is pinned down by the 90-10 difference in health spending (more poorly informed individuals leads to high variation in spending), and the covariance between health and age pins down the rate at which health depreciation increase with age, determining δ_{100} . Finally, δ_e is determined by the average difference in health between those who experience a health emergency in a given period and those who do not.